Expert Systems With Applications 315 (2026) 131634

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

A generalized zero-shot bearing fault diagnosis method under unseen faults and variable operating conditions

Jing Wang [a], Ning Li [a], Meng Zhou [a,*], Rong Su [b]

[a] School of Electrical and Control Engineering, North China University of Technology, Beijing, China
[b] School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore

ARTICLE INFO

Keywords:
Bearing fault diagnosis
Generalized zero-shot learning
Distribution discrepancy
Open-set gating
Domain shift

ABSTRACT

Bearings are essential components in mechanical systems, and accurate bearing fault diagnosis is critical for enabling intelligent manufacturing. Generalized zero-shot learning (GZSL) has emerged as a promising paradigm for recognizing unseen fault classes, but it remains limited to identify only specific unseen categories and is susceptible to domain shift. To address these challenges, this paper proposes a novel diagnostic framework, TransZeroNet. The core of the proposed framework lies in a distribution-discrepancy based gating mechanism. After wavelet transform, feature extraction, and alignment, samples from seen classes form compact class-specific distributions, whereas unseen classes deviate from the learned seen references. By quantifying such discrepancy, the proposed gate can reliably distinguish seen from unseen samples and accurately diagnose seen faults. For samples gated as unseen, an unsupervised clustering algorithm is further applied to achieve identification. Extensive experiments are conducted on three benchmark datasets-HEFEI, HUST and CWRU. Results demonstrate that TransZeroNet achieves high and stable recognition accuracies of 99.25%, 94.23%, and 99.50%, respectively, even under a few-shot training setting. These results verify that TransZeroNet is capable of handling both unseen faults and variable operating conditions.

1. Introduction

Bearings are the most critical components in rotating machinery, and they are often required to operate continuously under various heavy loads and in harsh environments (Luo et al., 2023). Bearing damage can lead to abnormal operating conditions, directly affect product quality and endanger the safety of operators. Bearing fault diagnosis methods are generally categorized into three types: physics-based approaches, qualitative knowledge-based approaches, and data-driven approaches. Physics-based methods rely on complete and accurate system modeling, which limits their generalizability. Qualitative knowledge-based methods are limited in their ability to cover all possible fault patterns. Data-driven methods, require large volumes of labeled or unlabeled samples (Ma et al., 2023). Its performance is typically constrained by the type, quantity and quality of training samples.

In practice, the test samples are often unpredictable and unobservable for training, which severely compromise the prediction accuracy and reliability of conventional deep models. Recent studies have termed the ability to detect such unseen classes as zero-shot fault diagnosis. In the GZSL fault diagnosis setting, the test set contains both seen and unseen categories, and the model is required to simultaneously maintain high recognition performance on seen faults while enabling knowledge transfer to unseen faults. (Huang et al., 2021). However, GZSL models typically suffer from a recognition bias, where predictions are skewed toward the seen classes due to higher confidence scores. To alleviate this bias, gated mechanism is proposed to determine whether a sample belongs to a seen or unseen class. It transforms the GZSL problem into a standard zero-shot problem to mitigate the recognition bias (Basu et al., 2023; Huang et al., 2023; Kömürcü & Petkevičius, 2024; Pan et al., 2022). The dual requirements of accurate recognition and bias mitigation make GZSL diagnosis both highly realistic and significantly challenging in industrial applications.

Zero-shot learning (ZSL) methods are generally divided into generative approaches (Liu et al., 2024; Xu et al., 2024a; Xu & Li, 2021), and embedding-based approaches (Chen et al., 2024a; Wu et al., 2025; Yang et al., 2025). ZSL method aims at extracting the shared knowledge between seen and unseen classes, and constructing a common semantic space to improve the generalization to unseen classes sample. However, this paradigm faces several challenges in bearing fault diagnosis. First, unlike the image tasks with features such as color and shape intuitively identifiable (Zhang & Zhang, 2024), commonalities between classes in bearing vibration signals are difficult to explicitly extract.

* Corresponding author.
E-mail addresses: jwang@ncut.edu.cn (J. Wang), li.n@mail.ncut.edu.cn (N. Li), zhoumeng@ncut.edu.cn (M. Zhou), RSu@ntu.edu.sg (R. Su).

https://doi.org/10.1016/j.eswa.2026.131634
Received 28 December 2025; Received in revised form 5 February 2026; Accepted 9 February 2026
Available online 13 February 2026

Moreover, class relationships are hard to model accurately whether in compound faults or signal variations under changing operating conditions. The modeling bias leads to misclassification when identifying real unseen classes using the models trained on synthetically generated data. So these models are typically limited to handling specific types of unseen faults or certain Variable operating conditions, which results in a poor robustness for diverse real environments.

To address the aforementioned challenges, this paper proposes a novel generalized zero-shot bearing fault diagnosis network, TransZeroNet. It is capable of simultaneously handling both unseen faults and variable operating condition diagnosis tasks under the GZSL setting. Cross-dataset zero-shot experiments were conducted on the HEFEI, HUST, and CWRU datasets to evaluate the performance of the proposed network. Experimental results demonstrate that TransZeroNet achieves the state-of-the-art accuracy with fewer parameters and efficient training, highlighting its effectiveness and practicality.

The main contributions of this paper are:

(1) TransZeroNet with hierarchy diagnosis scheme is proposed with four key modules aiming at the diagnosis of unseen faults and variable operating condition. Its highlight feature alignment module and hierarchy diagnosis are both designed based on the observed differences between seen and unseen classes.

(2) Feature alignment module quantifies the differences between seen and unseen classes using loss function-driven evaluation. The unseen samples are easily gated detected by class-gated recognition module which is training-free to enhance the robustness. The detected unseen class samples are further identified by unseen class recognition module with unsupervised learning strategy.

(3) TransZeroNet is verified on three typical bearing datasets to demonstrate its state-of-the-art performance both at the unseen fault diagnosis and varying operation scenarios under the generalized zero-shot setting. It eliminates the limitation of detecting only specific types of unseen faults compared with the exist GZSL diagnosis.

The remainder of this paper is organized as follows: Section 2 reviews the related work. Section 3 provides a detailed description of the proposed method. Section 4 presents comparative experiments and analysis. Finally, Section 5 draws a conclusion.

2. Related work

2.1. Zero-shot and generalized zero-shot fault diagnosis

Zero-shot learning (ZSL) and generalized zero-shot learning (GZSL) have become promising paradigms for fault diagnosis when collecting labeled data for all fault modes is impractical (Cen et al., 2025). Recently, generalized zero-shot learning bearing fault diagnosis (GZSL-BFD) has attracted increasing attention. Existing studies mainly focus on two major directions: (i) unseen faults diagnosis (Cen et al., 2025; Xu et al., 2023; Yang et al., 2025) and (ii) diagnosis under variable operating conditions (Chen et al., 2023; Liu et al., 2022; Long et al., 2024; Tang et al., 2025).

For unseen faults diagnosis, easily obtainable normal and single-fault samples are usually treated as seen classes, while compound faults are regarded as unseen classes. A compound fault refers to the simultaneous presence of two or more single faults. Due to fault coupling effects, compound faults often exhibit complex vibration patterns that are distinct from any individual fault type and may lead to more severe degradation. For the diagnosis under variable operating conditions, a certain operating condition (e.g., load/speed) is treated as the seen domain, and other conditions are considered unseen domains, making the task challenging due to strong distribution shift.

Many efforts have been made to recognize specific unseen fault types based on particular forms of auxiliary information. Honghuan Chen et al. fused the semantic construction and embedding-based method to enable GZSL diagnosis(Chen et al., 2024a). Jun Wang et al. introduced adaptive semantic-weighted auto-encoders to construct compound fault semantics for improved transferability (Wang et al., 2024). Yu Chen et al. incorporated an open-set recognition strategy to infer possible fault types of unseen samples(Chen et al., 2024b). Yi Qin et al. proposed the ZSAC model employing a multi-attribute label representation strategy for unseen fault diagnosis under the zero-shot setting (Qin et al., 2024). Jizheng Chen et al. applied a semantic-consistency generative adversarial network to handle zero-shot bearing fault diagnosis under variable operating conditions (Chen & Yang, 2024). Li Cai et al. developed a multi-attribute learning model specifically for GZSL fault diagnosis under variable operating conditions (Cai et al., 2024). Therefore, exploring zero-shot fault diagnosis under unseen faults and variable operating conditions is not only practically meaningful but also essential for modern bearing diagnostic systems deployed in complex and dynamic industrial scenarios.

2.2. Open-set recognition and OOD detection for reliable diagnosis

Open-set recognition (OSR) and out-of-distribution (OOD) detection have recently attracted increasing attention in intelligent fault diagnosis and industrial monitoring, since they improve reliability and trustworthiness under "unknown" conditions. Unlike conventional closed-set classification, OSR/OOD methods aim to detect test samples that do not belong to the training label space or deviate from the training data distribution, and then reject them as "unknown", thereby reducing overconfident misclassification in safety-critical industrial applications.

Representative OSR methods include OpenMax, which models classwise activation patterns and calibrates softmax outputs by fitting the tail distributions via Extreme Value Theory (EVT), producing an additional probability of the unknown class (Scheirer et al., 2011). In the OOD detection literature, widely-used baselines include Maximum Softmax Probability (MSP) and Energy-Based Scoring (EBO). Specifically, MSP rejects OOD samples by detecting confidence degradation, while EBO quantifies distribution discrepancy using an energy function derived from the logits (Liu et al., 2020). Although these techniques are effective for reliable unknown rejection, most OSR/OOD frameworks treat all unseen samples as a single unknown category and are not designed for fine-grained identification of unseen fault types. In other words, OSR/OOD primarily focuses on reliable rejection rather than unseen fault recognition.

2.3. Discussion: Differences between ZSL/GZSL and OSR/OOD

In summary, OSR and OOD detection mainly focus on rejecting samples outside the training label space, without providing a mechanism for fine-grained unseen fault identification. In contrast, GZSL aims at recognizing both seen and unseen categories by leveraging cross-class transfer. Our method follows the GZSL setting but introduces an OOD-style gating mechanism and an unseen-database-based clustering-and-voting strategy, enabling robust identification of unseen faults and variable operating conditions Table 1.

3. Methodology

3.1. Problem definition

In generalized zero-shot learning (GZSL) for bearing fault diagnosis, the fault label space is divided into two disjoint sets: the *seen* classes and the *unseen* classes. The training set contains labeled samples from the seen classes:

$$\mathcal{D}_s = \left\{(X_i, Y_i, R_i)\right\}_{i=1}^{n_s}, \quad Y_i \in \mathcal{Y}_s = \{0, 1, \dots, N_s - 1\} \tag{1}$$

where X_i denotes the ith vibration signal sample, Y_i is the corresponding class label, and R_i represents the corresponding reference feature used for gating. Here, n_s is the total number of training samples in the seen set, and $\mathcal{Y}_s$ denotes the label space of seen classes with $|\mathcal{Y}_s| = N_s$.

Table 1
Qualitative comparison of different methods.

Method Category	Semantic Dependency	Unseen-class Flexibility	Operating-condition Robustness
OOD detection (MSP, Energy, ODIN)	None	Low (reject-only)	Medium (depends on feature generalization)
OSR (OpenMax, ARPL)	None	Low-Medium (rejection, no fine-grained types)	Medium
Classical ZSL (semantic attribute based)	High	Medium (requires semantic vectors)	Low–Medium
Ours (GZSL + gating + unseen DB + voting)	**Low**	**High (unseen DB + clustering)**	**High (explicit alignment)**

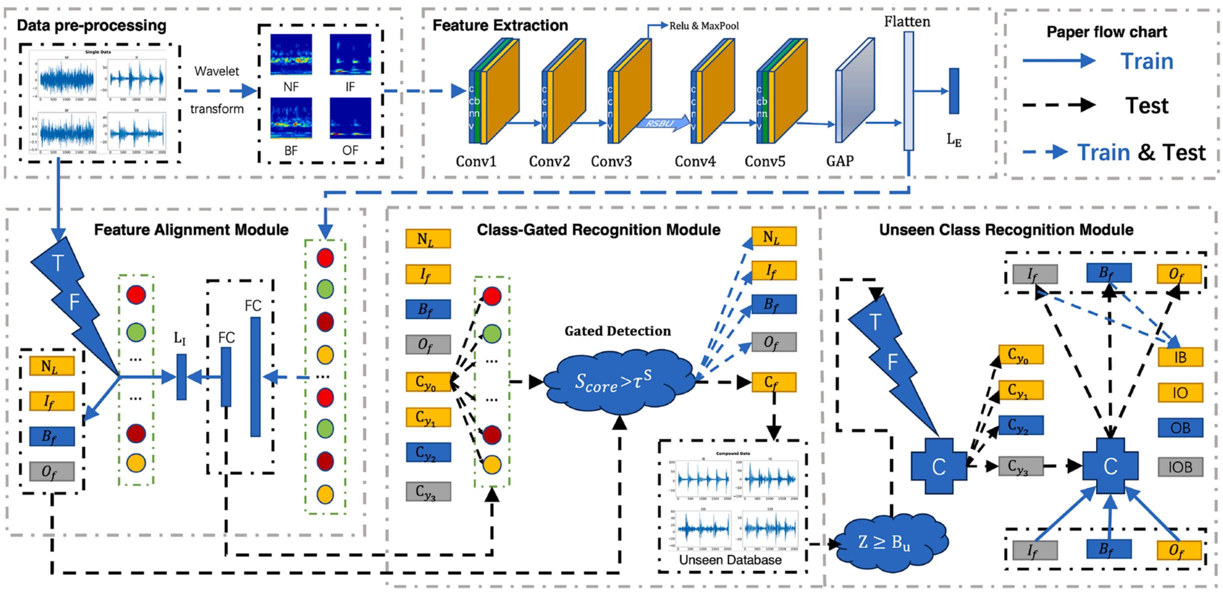


Fig. 1. General Framework Of TransZeroNet.

The unseen fault classes are unavailable during training. Let $\mathcal{Y}_u$ denote the label set of unseen classes with $|\mathcal{Y}_u| = N_u$. The seen and unseen sets satisfy $\mathcal{Y}_s \cap \mathcal{Y}_u = \emptyset$. The overall fault label space is defined as:

$$\mathcal{Y} = \mathcal{Y}_s \cup \mathcal{Y}_u = \{0, 1, \ldots, N_s - 1, \ldots, N_s + N_u - 1\}, \quad |\mathcal{Y}| = N_s + N_u \tag{2}$$

GZSL setting. The objective of GZSL is to train a diagnosis model $f(\cdot)$ using only samples from the seen classes:

$$f : \mathcal{X} \to \mathcal{Y}, \quad \text{trained on } \mathcal{D}_s \tag{3}$$

and evaluate it on test samples that may come from both seen and unseen classes:

$$\mathcal{D}_{test} \subseteq \mathcal{Y}_s \cup \mathcal{Y}_u \tag{4}$$

Unlike conventional zero-shot learning (ZSL), where the test set only contains unseen classes ($\mathcal{D}_{test} \subseteq \mathcal{Y}_u$), the GZSL formulation better reflects real industrial scenarios, in which both seen and novel fault types may occur during deployment.

3.2. TransZeroNet architecture

TransZeroNet leverages the differences between distinct signals and the similarities within the same signal to construct a robust recognition mechanism. As illustrated in Fig. 1, the model consists of five main modules: Data Pre-processing Module, Feature Extraction Module, Feature Alignment Module, Class-Gated Recognition Module and Unseen Class Recognition Module.

3.2.1. Data pre-processing module

To preserve the characteristics of raw one-dimensional signals $f(t)$ across both time and frequency domains, Continuous Wavelet Transform (CWT) is adopted for data preprocessing in this study. The resulting time-frequency images contain rich information (Zhang & Yan, 2006). CWT transformation is formally defined as,

$$WT(a, \tau) = \int_{-\infty}^{+\infty} f(t) \frac{1}{\sqrt{a}} e^{-\frac{(t-\tau)^2}{2a^2}} \cos\left(\frac{5(t-\tau)}{a}\right) dt \tag{5}$$

were $a > 0$ and τ are the frequency scale factor and the time shift parameter, respectively. The mother wavelet function is Morlet wavelet defined as $\psi(t) = e^{-\frac{t^2}{2}} \cos(5t)$. Then raw bearing signals are transformed into $3 * 64 * 64$ wavelet time-frequency images.

3.2.2. Feature extraction module

The backbone of feature extraction module primarily consists of five convolutional layers (Conv), one RSBU-CW (Residual Shrinkage Building Units with Channel-Wise attention), a Global Average Pooling (GAP) layer, a Flatten layer, and a Fully Connected (FC) layer. The preprocessed $3 * 64 * 64$ time-frequency images are fed into the backbone network for training and feature extraction. The detailed architecture is illustrated in Table 2.

It is noted that the RSBU-CW module is integrated, which helps suppress image noise and alleviate the gradient vanishing problem (Bai et al., 2022; Fang et al., 2024; Wang et al., 2023; Xu et al., 2024b). It consists of two convolutional layers and a Get Threshold submodule, illustrated in Fig. 2. The input of RSBU is the latent feature vector X_I. It is first passed through two convolutional layers to produce the intermediate output X_C which is fed into the Get Threshold submodule. Here X_C is subjected to absolute value transformation followed by Global Average Pooling (GAP) to generate X_r. Then X_r is passed through two fully connected (FC) layers to obtain X_B, and subsequently activated by Sigmoid function. Then the corresponding scaling factor β is obtained

Table 2
Structure parameters of feature extractor module.

Structural Layer	Conv Kernel	Padding	Maxpooling Kernel	Output Channel	Feature Size
Input	–	–	–	3	3 ∗ 64 ∗ 64
Conv1	3*3	1	2*2	64	64 ∗ 32 ∗ 32
Conv2	3*3	1	2*2	64	64 ∗ 16 ∗ 16
Conv3	3*3	2	2*2	256	256 ∗ 9 ∗ 9
RSBU-CW	–	–	–	–	256 ∗ 9 ∗ 9
Conv4	3*3	2	2*2	512	512 ∗ 5 ∗ 5
Conv5	3*3	2	2*2	1024	1024 ∗ 3 ∗ 3
GAP	–	–	–	–	1024 ∗ 1 ∗ 1
Flatten	–	–	–	–	1024
Dropout1d	0.5	–	–	–	1024
FC	–	–	–	4	4

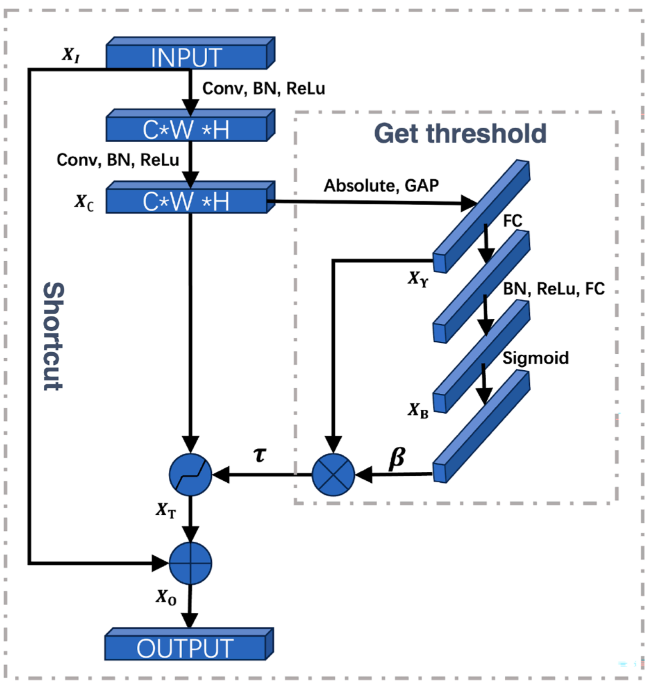


Fig. 2. RSBU-CW structure.

with range (0,1),

$$
\begin{aligned}
X_C &= Relu(BN(Conv(Relu(BN(Conv(X_I)))))) \\
X_r &= \mathrm{GAP}\left(\left|X_C\right|\right) \\
X_B &= \mathrm{FC}\left(\mathrm{ReLu}\left(\mathrm{BN}\left(FC\left(X_r\right)\right)\right)\right) \\
\beta &= e^{X_B} / \left(1 + e^{X_B}\right)
\end{aligned}
\tag{6}
$$

Each channel within RSBU-CW block has its own individual threshold, enhancing the network's denoising capability. The soft threshold τ_c of the cth channel is,

$$
\tau_c = \beta \cdot X_r \tag{7}
$$

Then X_T is obtained through a piecewise function, and combined with input X_I via a shortcut connection to produce the final output vector X_O,

$$
\begin{aligned}
X_T &= \begin{cases} X_C - \tau, & \left(X_C > \tau\right) \\ 0, & \left(-\tau < X_C < \tau\right) \\ X_C + \tau, & \left(X_C < -\tau\right) \end{cases} \\
X_O &= X_I + X_T
\end{aligned}
\tag{8}
$$

Feature extraction is trained with Cross Entropy Loss,

$$
\mathcal{L}_{\text{extraction}} = -\log\left(\frac{e^{X_S}}{\sum_{i=1}^{N_S} e^{X_i}}\right) \tag{9}
$$

Specifically e^{X_S} is the exponential of the logit output corresponding to seen class S, $\sum_{i=1}^{N_s} e^{X_i}$ is the sum of the logit exponentials over all N_S seen classes. The output of Flatten layer is regarded as the extracted deep feature $\hat{x}_i$,

$$
\hat{x}_i = \text{Extraction}\left(x_i, \theta\right) \tag{10}
$$

where x_i is the wavelet time-frequency images.

3.2.3. *Feature alignment module*

The Feature Alignment Module consists of a Time & Frequency (TF) Module and an Alignment Module.

TF module calculates the 28-dimensional time-frequency statistical features from one-dimensional bearing signals of seen classes. It includes 15 time-domain and 13 frequency-domain indicators detailed in Table 3. The time-frequency reference feature vector is constructed,

$$
\mathrm{R}^S = \left\{ \left(r_1, \ldots, r_k, \ldots, r_{28}\right)_S^{\mathcal{Y}_s} \mid S \in \mathcal{Y}_s \right\} \tag{11}
$$

Table 3
Time & frequency statistical features.

Calculation formulas for time-domain statistical indicators.	
$a_1 = \frac{1}{N} \sum_{n=1}^{N} x(n)$	$a_9 = \min(x_n)$
$a_2 = \frac{1}{N} \sum_{n=1}^{N} \lvert x(n)\rvert$	$a_{10} = t_6/t_1$
$a_3 = \frac{1}{N} \sum_{n=1}^{N} x^2(n)$	$a_{11} = t_7/t_6$
$a_4 = \sqrt{\frac{1}{N-1} \sum_{n=1}^{N} \left[x(n) - t_1\right]^2}$	$a_{12} = t_7/t_1$
$a_5 = \left(\frac{1}{N} \sum_{n=1}^{N} \sqrt{\lvert x(n)\rvert}\right)^2$	$a_{13} = t_7/t_5$
$a_6 = \sqrt{\frac{1}{N} \sum_{n=1}^{N} x^2(n)}$	$a_{14} = \frac{\sum_{n=1}^{N} [x(n)-t_1]^3}{(N-1)t_4^3}$
$a_7 = \max \lvert x(n)\rvert$	$a_{15} = \frac{\sum_{n=1}^{N} [x(n)-t_1]^4}{(N-1)t_4^4}$
$a_8 = \max(x_n)$	
Calculation formulas for frequency domain statistical indicators.	
$a_{16} = \frac{1}{K} \sum_{k=1}^{K} s(k)$	$a_{23} = \sqrt{\frac{\sum_{k=1}^{K} f_k^4 \cdot s(k)}{\sum_{k=1}^{K} f_k^2 \cdot s(k)}}$
$a_{17} = \sqrt{\frac{1}{K-1} \sum_{k=1}^{K} [s(k) - t_{16}]^2}$	$a_{24} = \frac{\sum_{k=1}^{K} f_k^2 \cdot s(k)}{\sqrt{\sum_{k=1}^{K} s(k) \cdot \sum_{k=1}^{K} f_k^4 \cdot s(k)}}$
$a_{18} = \frac{\sum_{k=1}^{K} [s(k)-t_{16}]^3}{(K-1)t_{17}^3}$	$a_{25} = t_{22}/t_{20}$
$a_{19} = \frac{\sum_{k=1}^{K} [s(k)-t_{16}]^4}{(K-1)t_{17}^4}$	$a_{26} = \frac{\sum_{k=1}^{K} (f_k-t_{20})^3 \cdot s(k)}{(K-1)t_{22}^3}$
$a_{20} = \frac{\sum_{k=1}^{K} f_k \cdot s(k)}{\sum_{k=1}^{K} s(k)}$	$a_{27} = \frac{\sum_{k=1}^{K} (f_k-t_{20})^4 \cdot s(k)}{(K-1)t_{22}^4}$
$a_{21} = \sqrt{\frac{\sum_{k=1}^{K} f_k^2 \cdot s(k)}{\sum_{k=1}^{K} s(k)}}$	$a_{28} = \frac{\sum_{k=1}^{K} (f_k-t_{20})^{1/2} \cdot s(k)}{(K-1)t_{22}^{1/2}}$
$a_{22} = \sqrt{\frac{1}{K-1} \sum_{k=1}^{K} (f_k - t_{20})^2 \cdot s(k)}$	

Note: $x(n)$: time domain sequence , $s(k)$: frequency spectrum.

Table 4
Structure parameter of feature alignment module.

Layer	Operator	Kernel Size
Input	Input Signal	1024
FC1	Fully Connected	1024*256
Lrelu	Leaky Relu	0.2
Dropout1d	–	0.5
FC2	Fully Connected	256*28

where S is the Sth seen class in $\mathcal{Y}_s$. Subsequently, feature-wise normalization is performed,

$$\hat{r}_k = \frac{r_k - \min(r_k)}{\max(r_k) - \min(r_k)} \tag{12}$$

Then we obtain the normalized time-frequency reference features $\hat{r}^S$ with a shape of $N_s * 28$.

The Feature Alignment Module primarily consists of two fully connected (FC) layers with the detailed architecture in Table 4. It encourages the deep feature $\hat{x}_i$ to be distribution-consistent with the normalized time-frequency reference feature extracted by the TF module, thereby aligning seen-class representations with their corresponding reference features. A similarity-based loss is employed to optimize this alignment,

$$\mathcal{L}_{\text{alignment}} = 1 - \frac{1}{N}\sum_{i=1}^{N} \frac{\mathbf{p}_i \cdot \hat{r}^S}{\|\mathbf{p}_i\| \|\hat{r}^S\|} \tag{13}$$

where $\mathbf{p}_i = \text{Alignment}(\hat{x}_i)$, N is the number of samples in a batch. $\hat{x}_i$ is the deep feature of seen classes S, and $\hat{r}^S$ is the normalized time-frequency reference features of class S.

During validation, all deep features $\hat{x}_i^{val}$ are fed into the feature alignment module for evaluation, producing the aligned reference features $\hat{R}^{S_{\text{val}}} \in \mathbb{R}^{\text{Len}(val)\times 28}$.

$$\begin{aligned}\hat{R}^{S_{\text{val}}} &= W_2\left(\text{LeakyReLU}\left(W_1 \cdot \hat{x}_i^{\text{val}} + b_1\right) + b_2\right) \\ &= \left\{ \left(r_1^S, \dots, r_k^S, \dots, r_{28}^S\right)_{i=1}^{\text{Len(val)}} \,\middle|\, S \in \mathcal{Y}_s \right\}\end{aligned} \tag{14}$$

Here W_1, W_2, b_1 and b_2 represent the weight matrices and bias vectors of the first and second fully connected layers in the Alignment Module, respectively. $\mathcal{Y}_s$ denotes the label set of the seen classes. Meanwhile, key performance indicators (KPIs) are collected on the validation set. In this work, the KPI is defined as the similarity score, which corresponds to the output of the similarity-based objective.

$$\text{KPI}^{\text{val}} = \frac{\left(\hat{R}_i^{S_{val}}\right)_{i=1}^{\text{Len}(val)} \cdot \hat{r}^S}{\left\|\left(\hat{R}_i^{S_{val}}\right)_{i=1}^{\text{Len}\,(val)}\right\| \|\hat{r}^S\|} \tag{15}$$

To set a class-specific gating threshold, we compute the KPI scores of validation samples belonging to each seen class. Since the KPI is defined as a similarity-based score, a higher value indicates that the sample is more consistent with the normalized reference feature $\hat{r}^S$ of the corresponding seen class, whereas a lower value implies a larger distributional deviation. Therefore, within the validation set, the minimum KPI value represents the most challenging (i.e., least similar) seen-class sample that is still considered in-distribution. By defining the gating threshold as

$$\tau^S = \text{Min}\left(KPI^{val}\right) \tag{16}$$

the decision rule ensures that all validation samples from this seen class satisfy $\text{Score} \geq \tau^S$, providing a conservative boundary that preserves seen-class recall while rejecting samples whose distribution deviates beyond the observed seen variation.

3.2.4. Class-gated recognition module

During the testing phase, both seen and unseen samples are fed into the trained network. Since the feature alignment module constrains seen-class samples to follow a compact reference distribution, unseen samples tend to exhibit larger distribution discrepancy in the learned representation space. By computing the quantification scores (KPI^S) and applying the optimized thresholds τ^S learned for each seen class, the gating unit can effectively distinguish seen from unseen samples, and further assign seen samples to their corresponding seen fault classes.

Specifically, given test deep features $\hat{x}_i^{test}$ with batch size B, the alignment module maps them into aligned reference features $\hat{R}^{S_{\text{test}}} \in \mathbb{R}^{B\times 28}$.

$$\hat{R}^{\text{test}} = W_2\left(\text{LeakyReLU}\left(W_1 \cdot \hat{x}_i^{\text{test}} + b_1\right) + b_2\right) \tag{17}$$

The main process is summarized as follows:

1. For each test sample, the cosine similarity is computed between its aligned reference feature $\hat{R}_i^{\text{test}}$ ($i = 1, 2, \dots, B$) and the seen-class normalized reference features $\hat{r}^S$ ($S = 1, 2, \dots, N_s$). This yields a similarity (KPI) matrix $\mathbf{KPI} \in \mathbb{R}^{B\times N_s}$:

$$\text{KPI}^{\text{test}} = \frac{\left(\hat{R}_i^{\text{test}}\right)_{i=1}^{B} \cdot \hat{r}^S}{\left\|\left(\hat{R}_i^{\text{test}}\right)_{i=1}^{B}\right\| \|\hat{r}^S\|} \tag{18}$$

2. For each test sample, the maximum similarity score and the corresponding matched seen class index are obtained:

$$\text{Score}_i = \max_{s\in\{1,\dots,N_s\}} \text{KPI}_{i,s}^{test}, \quad \text{Index}_i = \arg\max_{s\in\{1,\dots,N_s\}} \text{KPI}_{i,s}^{test}. \tag{19}$$

3. The obtained Index_i corresponds to a seen class S, whose threshold τ^S is retrieved. The gating decision is then made as:

$$\text{Gated Detection} = \begin{cases} \text{Seen}, & \text{if } \text{Score}_i \geq \tau^S \\ \text{Unseen}, & \text{otherwise} \end{cases} \tag{20}$$

If Gated Detection = Seen, the test sample is identified as belonging to a seen class, and the corresponding class S is assigned as its predicted label. If Gated Detection = Unseen, the test sample is regarded as an unseen-fault candidate, and its one-dimensional bearing signal is appended to an *Unseen Database* for subsequent unseen-fault identification.

In practical industrial monitoring, bearing vibration signals are acquired in an online and continuous manner. Therefore, an *Unseen Database* is designed to temporarily store the gated unseen samples, enabling subsequent unseen-fault inference and identification.

To satisfy real-time requirements, a lightweight triggering mechanism is adopted. Specifically, the *Unseen Class Recognition Module* is activated once the number of buffered unseen candidates reaches a predefined threshold B_u. Let Z denote the current number of samples stored in the Unseen Database. The trigger rule is defined as:

$$\text{Trigger} = \begin{cases} \text{True}, & Z \geq B_u \\ \text{False}, & \text{otherwise} \end{cases} \tag{21}$$

This design enables timely unseen-fault identification after alarm triggering, while maintaining a limited recognition delay. Once triggered, the buffered unseen candidates in the *Unseen Database* are further analyzed by the *Unseen Class Recognition Module*.

3.2.5. Unseen class recognition module

Once the trigger condition in Eq. (21) is satisfied, the samples stored in the *Unseen Database* are processed for unseen-fault inference. First, each gated unseen signal is fed into the TF module to extract its time-frequency reference feature vector R_i^u. The resulting feature set $\{R_i^u\}_{i=1}^{Z}$ is then clustered using K-means to discover potential unseen fault patterns:

$$\{\ell_i\}_{i=1}^{Z} = \text{Kmeans}\left(\{R_i^u\}_{i=1}^{Z}, K_c\right) \tag{22}$$

where $\ell_i \in \{1, 2, \dots, K_c\}$ denotes the cluster assignment of the ith gated unseen sample, and Z denotes the number of buffered samples stored in the Unseen Database. Accordingly, the mth cluster is denoted as $\mathcal{K}_m = \{i \mid \ell_i = m\}$.

Assuming that unseen fault patterns may exhibit latent similarity with seen fault categories in the reference feature space, the clustered abnormal patterns are further aligned with seen-class prototypes for pseudo-label assignment. Specifically, the reference features $\{R_i^S\}_{i=1}^{n_s}$ are extracted from the seen fault training samples using the TF module, and a discriminative projection function $\phi(\cdot)$ is learned using the Large Margin Nearest Neighbor (LMNN) algorithm:

$$\phi(\cdot) = \mathrm{LMNN}\left(\{R_i^S\}_{i=1}^{n_s}, \{Y_i^S\}_{i=1}^{n_s}\right) \tag{23}$$

where $\phi(\cdot)$ maps reference features into a discriminative seen-fault embedding space. Subsequently, each gated unseen sample is projected via $\phi(\cdot)$, and a KNN classifier is applied in the LMNN space to obtain a pseudo label:

$$\hat{c}_i = \mathrm{KNN}\left(\phi(R_i^u)\right), \quad i = 1, 2, \dots, Z \tag{24}$$

where $\hat{c}_i \in \{1, 2, \dots, N_s\}$ indicates the predicted seen-class label for the ith unseen sample. For each cluster $\mathcal{K}_m$, a majority vote is conducted over the pseudo labels of its member samples:

$$v_m(c) = \sum_{i \in \mathcal{K}_m} \mathbb{1}(\hat{c}_i = c) \tag{25}$$

where $\mathbb{1}(\cdot)$ is the indicator function and $v_m(c)$ counts the occurrences of class c in cluster $\mathcal{K}_m$. Finally, the cluster-level label is determined as:

$$\hat{y}_m = \begin{cases} \arg\max_{c \in C} v_m(c), & \frac{\max_c v_m(c)}{|\mathcal{K}_m|} \geq \tau_{\text{unseen}} \\ \text{Manual annotation}, & \text{otherwise} \end{cases} \tag{26}$$

where $C = \{1, 2, \dots, N_s\}$ denotes the seen-class label space, $|\mathcal{K}_m|$ is the number of samples in cluster $\mathcal{K}_m$, and τ_{unseen} is a confidence threshold (set to 0.6–0.7 in this work). The detailed algorithmic procedure is provided in Algorithm 1.

Algorithm 1 Unseen fault identification with unseen database and cluster-wise voting.

Require: Buffered unseen signals $\{x_i^u\}_{i=1}^{Z}$, TF extractor $\mathcal{F}_{TF}(\cdot)$, number of clusters B_u, LMNN projection $\phi(\cdot)$ trained on seen reference features, KNN classifier in LMNN space, confidence threshold τ_{unseen}

Ensure: Cluster-level unseen labels $\{\hat{y}_m\}_{m=1}^{K_c}$ or manual annotation

```
Extract reference features for buffered unseen samples:
    R_i^u ← F_TF(x_i^u),   i = 1, ..., Z.
Perform clustering in reference feature space:
    {ℓ_i}_{i=1}^Z ← Kmeans({R_i^u}_{i=1}^Z, K_c).
Construct clusters:
    K_m ← {i | ℓ_i = m},   m = 1, ..., K_c.
for i = 1 to Z do
  Obtain pseudo label for each buffered sample:
    ĉ_i ← KNN(φ(R_i^u)).
end for
for m = 1 to K_c do
  Majority voting within cluster K_m:
    v_m(c) ← Σ_{i∈K_m} 1(ĉ_i = c).
  Compute voting confidence:
    T_m ← max_{c∈C} v_m(c),   Conf_m ← T_m/|K_m|.
  if Conf_m ≥ τ_unseen then
    Assign cluster-level label:
      ŷ_m ← arg max_{c∈C} v_m(c).
  else
    ŷ_m ← Manual annotation.
  end if
end for
return {ŷ_m}_{m=1}^{K_c}
```

4. Experiment and discussion

4.1. Experimental settings

To evaluate the performance of the proposed model across different datasets, a series of experiments were conducted under both unseen faults and variable operating conditions scenarios on the HEFEI (Xu & Li, 2021), HUST (Thuan & Hong, 2023), and CWRU (Hendriks et al., 2022) datasets. The experimental settings are detailed in Table 5.

HEFEI & HUST Datasets: Unseen fault experiments were conducted on both datasets. Each dataset includes bearing conditions: Healthy (H), Inner Race Fault (IF), Ball Fault (BF), Outer Race Fault (OF), and compound faults-IB, IO, OB, and IOB (IOB only in Hefei). H, IF, BF, and OF are used as seen classes for training/testing; the compound faults are unseen classes and used only during testing. The HUST dataset used in this study involves bearing type 6208 under 400 W load.

Table 5
Experimental environment setting.

Name	Version
Hardware	M2 Pro, 16GB, MPS:0
Software	Pycharm2023.2.7
Programming Language	Python version: 3.11.9
Model Framework	Pytorch2.2.2, Torchvision0.17.2
Random Seed	42
NumPy	1.26.4
Pandas	2.2.2
Matplotlib	3.9.1
Train Size	0.8
Val-Size	0.2

CWRU Dataset: Variable operating experiments were conducted on CWRU dataset. It includes bearing conditions under different operating scenarios: Healthy (H), Inner Race Fault (IF), Ball Fault (BF), and Outer Race Fault (OF).

The evaluation metrics include S_{acc} and U_{acc}, denoting accuracies on seen and unseen classes, respectively. H_{acc} offers a reliable assessment of the model's generalization (Lee et al., 2023),

$$H_{\text{acc}} = 2 \times S_{\text{acc}} \times U_{\text{acc}} / \left(S_{\text{acc}} + U_{\text{acc}}\right) \tag{27}$$

In addition, since our framework employs a gating module to distinguish seen/unseen samples, we further evaluate the gate performance using three metrics: FPR^{seen}, TPR^{unseen}, and $AUROC^{Gated}$. Here, unseen samples are treated as the positive class, while seen samples are treated as the negative class. Accordingly, TP and FN denote the numbers of unseen samples correctly gated as unseen and incorrectly gated as seen, respectively. Meanwhile, FP and TN denote the numbers of seen samples incorrectly gated as unseen and correctly gated as seen, respectively. Based on these definitions, FPR^{seen}, TPR^{unseen}, and $AUROC^{Gated}$ are defined as:

$$FPR^{seen} = \frac{FP}{FP + TN}, \qquad TPR^{unseen} = \frac{TP}{TP + FN} \tag{28}$$

$$AUROC^{Gated} = \int_0^1 TPR^{unseen}(FPR^{seen})\, dFPR^{seen} \tag{29}$$

4.2. Experimental results analysis

Identifying normal and faulty samples is a prerequisite for effective fault diagnosis. To evaluate the performance of the proposed model's class-gated recognition module across different datasets, the confusion matrices of the detection results are shown in Fig. 3.

The proposed model achieved 100% seen and unseen fault detection rate across all datasets, indicating that TransZeroNet can provide reliable monitoring of equipment operational status.

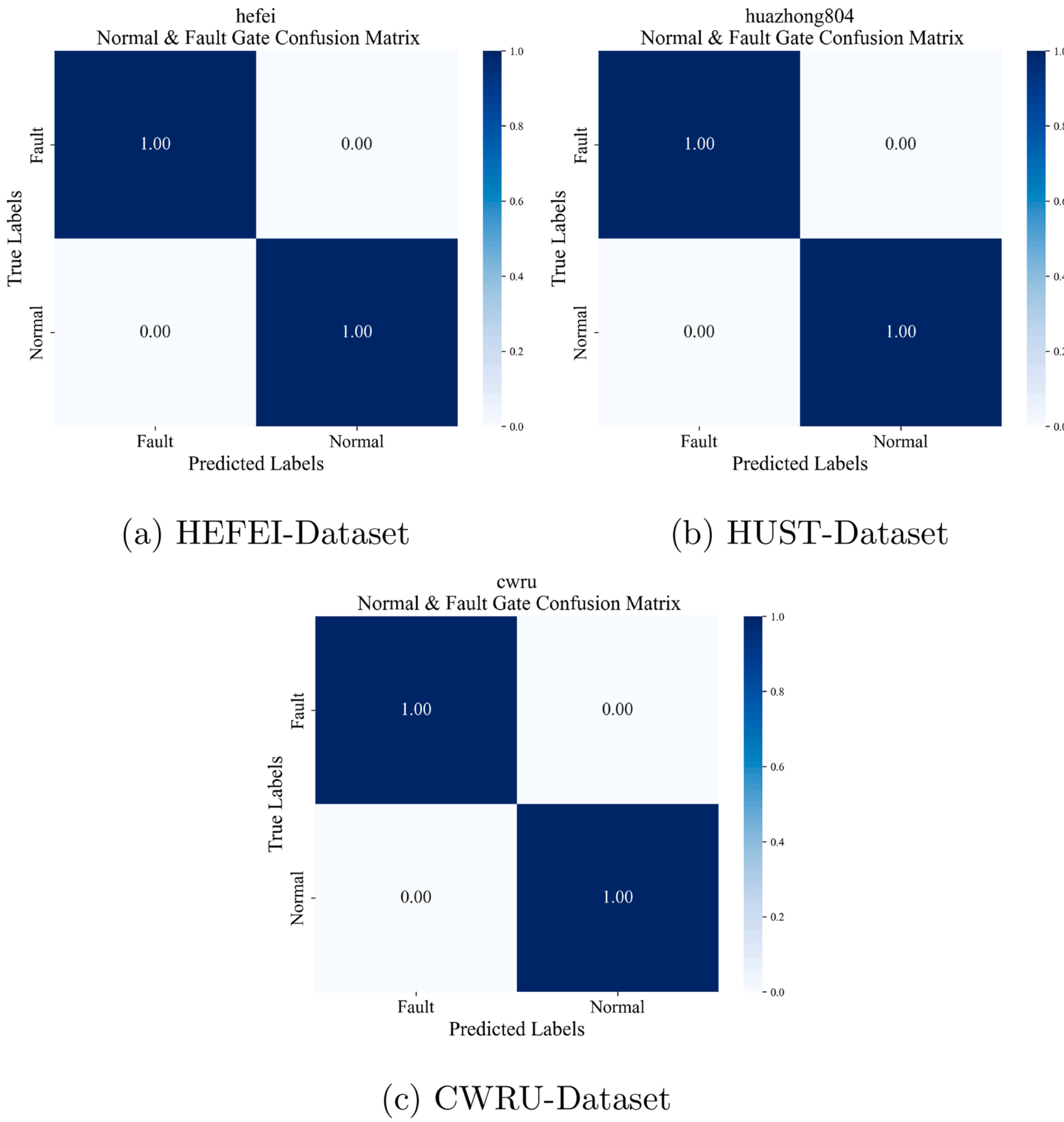


Fig. 3. Normal Fault Detection.

4.2.1. *Effect of pre-processing methods and image resolution*

To assess the effects of preprocessing and image resolution on model performance, comparative experiments were conducted under consistent settings. Here the one-dimensional signal is conversed into image. Three conversion algorithms (GASF, GADF, and CWT) and three resolutions (64×64, 128×128, and 256×256) were evaluated with H_{acc} as the metric. Results are shown in Table 7, with parameter details in Table 6.

The results indicate that increasing the size of feature map within a reasonable range enhances performance. However excessively large sizes may introduce redundancy and higher computational cost. Among them, CWT outperforms GASF and GADF in most cases due to its richer time-frequency representations. Considering both accuracy and efficiency, the $64 * 64$ CWT image offers an optimal trade-off and is thus adopted for data preprocessing.

4.2.2. *Structure and parameter tuning*

In Table 8, Gated_{normal}, Gated_{fault}, Gated_{seen}, and Gated_{unseen} represent the recognition accuracy of normal samples, fault samples, seen-class samples, and unseen-class samples by the gated network, respectively.

From the results, it can be observed that under the proposed framework, the proposed feature extraction module significantly outperforms CNN-Transformer and ResNet variants. On the HEFEI dataset, the gated module integrated with the proposed feature extraction module is able to effectively recognize both seen and unseen classes, achieving an overall harmonic accuracy of 99.25%.

It is further observed that increasing the number of convolutional layers, fully connected layers, or RSBU-CW modules does not yield notable improvements in the recognition performance of the gated module. On the contrary, enlarging the model may result in prolonged training times, increased risk of overfitting, and, in certain cases, a decline in overall performance. For instance, the introduction of dropout within convolutional layers can cause partial information loss, which adversely affects the network's effectiveness. These findings suggest that the network architecture must be carefully designed and appropriately balanced to optimize both feature extraction and feature alignment capabilities.

Since the training stage only involves the feature extraction module and the feature alignment module, whereas the inference-stage decision is mainly dominated by the proposed class-gated recognition module

Table 6
Model Parameters.

Module	Parameter	HEIFEI	HUST	CWRU
Dataset Maker	silding_window	2048	2048	2048
	overlap_ratio	0.6	0.8	0.8
	train_size	298	578	195
	val_size	75	145	49
	test_size	250	100	50
Wavelet Transform	sampling_period	1.0/51200	1.0/51200	1.0/51200
	totalscal	128	128	128
	wavename	cmor3-3	cmor3-3	cmor3-3
	image_size	64	256	64
Extraction Module	batch_size	32	32	32
	learn_rate	1e-4	1e-4	1e-4
	epoches_number	15	15	15
Alignment Module	batch_size	64	64	64
	learn_rate	1e-5	1e-5	1e-5
	epoches_number	120	120	120
TF Module	Fs	51,200	51,200	51,200
K-means Cluster	epoches_number	300	300	300
	Number of clusters	$N_{seen}+1$	$N_{seen}+1$	$N_{seen}+1$
	Distance metric	Euclidean	Euclidean	Euclidean
	n_init	10	10	10

Table 7
H_{acc} Under different pre-processing.

Dataset	Method	64*64	128*128	256*256
HEFEI	GASF	52.95%	58.94%	46.99%
	GADF	49.27%	60.92%	80.12%
	CWT	**99.25%**	**98.38%**	**93.88%**
HUST	GASF	38.99%	53.32%	65.36%
	GADF	79.08%	68.77%	67.59%
	CWT	**94.23%**	**91.89%**	**94.21%**

and unseen-class identification module, the overall recognition mechanism differs from conventional end-to-end deep classifiers. In particular, our training objective focuses on enforcing consistency between seen-class samples and their corresponding normalized reference features (Fig. 4), rather than maximizing closed-set classification accuracy on all fault categories. As a result, the near-100% performance in some metrics does not necessarily indicate overfitting. Instead, it reflects the high separability of the gating task (e.g., normal vs. abnormal, or seen vs. unseen), which is explicitly quantified by the proposed distribution discrepancy score. Moreover, the training and validation curves of the extraction and alignment modules (Fig. 5) exhibit smooth convergence and stable optimization without obvious divergence between training and validation trends, which indirectly suggests that the proposed framework does not suffer from typical overfitting issues.

4.2.3. Experiments on KPI functions

To justify the choice of the optimization objective in the proposed feature alignment module, we conduct a comparative study between similarity-based and distance-based learning objectives. Specifically, the alignment module aims to enforce consistency between the extracted deep feature of a seen-class sample and its corresponding normalized reference feature vector. Therefore, different quantification functions may lead to significantly different alignment behaviors and gating performance.

As reported in Table 9, the similarity-based objective consistently outperforms the distance-based alternative on both HEFEI and HUST datasets. This phenomenon can be explained as follows. The alignment module is implemented as a lightweight network with only shallow fully connected layers, which makes it difficult to reliably minimize absolute Euclidean distance in the feature space, especially under operating-condition variations and distribution shifts. As a result, distance-based alignment tends to produce unstable convergence and degraded generalization.

In contrast, similarity-based objectives (e.g., cosine similarity loss) optimize directional consistency at the vector level, encouraging the aligned feature to be highly consistent with the seen-class reference vector, i.e., driving their similarity toward one. This property matches the design goal of the proposed gating score, and leads to more separable KPI distributions between seen and unseen samples. Consequently, the discriminative capability of the Class-Gated Recognition Module is significantly improved. The qualitative distributions and classification results under different quantification metrics are further visualized in Fig. 6.

Fig. 6a and b illustrate the distributions of the KPI scores under similarity-based and distance-based quantification, respectively. In particular, in Fig. 6b, the KPI scores of seen and unseen samples exhibit significant overlap, which reduces the discriminative capability of the Class-Gated Recognition Module. Fig. 6c and d further confirm that similarity-based metrics yield better discrimination, reducing misclassification of seen as unseen classes. Therefore, similarity is adopted as the quantitative indicator due to its superior separability and recognition performance.

Fig. 6a and b illustrate the distributions of KPI scores under similarity-based and distance-based quantification, respectively. As shown in Fig. 6b, the KPI distributions of seen and unseen samples are heavily overlapped, which weakens the separability and thus degrades the discriminative capability of the Class-Gated Recognition Module. Fig. 6c and d further confirm that similarity-based quantification yields better seen/unseen discrimination, significantly reducing the misclassification between seen and unseen classes. Therefore, cosine similarity is adopted as the KPI function due to its superior separability and recognition performance.

4.2.4. Sensitivity analysis of thresholding strategies

Table 10 reports the gate performance under different thresholding strategies, including percentile-based thresholds, trimmed-min thresholds, EVT-based tail modeling, and FPR-targeted thresholds. Overall, the proposed gating module remains highly stable across different thresholding methods, achieving $AUROC^{Gated} \approx 0.9995$ and 1.0000 on the HEFEI and CWRU datasets, respectively. This observation indicates that the proposed distribution-discrepancy based gate is robust and does not critically rely on a particular thresholding formulation.

Table 8
Structure experiment (train modules).

Module	Structure	$Gated_{normal}$	$Gated_{fault}$	$Gated_{seen}$	$Gated_{unseen}$	S_{Acc}	U_{Acc}	H_{Acc}
TransZeroNet	–	100.00%	100.00%	100.00%	100.00%	99.70%	98.80%	99.25%
Extraction Module	CNN-Transformer	100.00%	100.00%	99.00%	83.00%	98.80%	81.40%	89.26%
	Resent-18	99.00%	100.00%	96.60%	99.00%	97.00%	95.10%	97.06%
	Resent-34	100.00%	100.00%	99.00%	87.00%	98.90%	85.80%	91.89%
	Resent-50	99.00%	100.00%	99.00%	97.00%	99.20%	95.90%	97.52%
	Resent-101	99.00%	100.00%	97.00%	92.00%	99.20%	70.60%	82.49%
	Resent-152	99.00%	100.00%	98.00%	95.00%	99.40%	71.60%	83.24%
Extraction Module	ConvLayer −1	100.00%	100.00%	100.00%	95.00%	99.60%	93.40%	96.40%
	ConvLayer +1	99.00%	100.00%	100.00%	100.00%	99.60%	98.60%	99.10%
	Dropout -1D	100.00%	100.00%	99.00%	100.00%	99.50%	98.70%	99.10%
	Dropout +2D	100.00%	100.00%	99.00%	99.00%	99.50%	98.20%	98.85%
	FullConnLayer +1	100.00%	100.00%	99.00%	100.00%	99.50%	74.80%	85.40%
	RSBU-CW −1	99.00%	100.00%	100.00%	100.00%	99.60%	98.40%	99.00%
	RSBU-CW +1	100.00%	100.00%	100.00%	95.00%	99.60%	94.40%	96.93%
	RSBU-CW +2	100.00%	100.00%	100.00%	100.00%	**99.90%**	96.10%	97.96%
Alignment Module	FullConnLayer −1	100.00%	100.00%	100.00%	99.00%	**99.80%**	98.00%	98.89%
	FullConnLayer +1	100.00%	100.00%	100.00%	100.00%	99.70%	98.40	99.05%
	Droupout −1	100.00%	100.00%	100.00%	100.00%	99.70%	98.10%	98.89%
	Droupout +1	100.00%	100.00%	100.00%	100.00%	74.70%	85.41%	98.89%

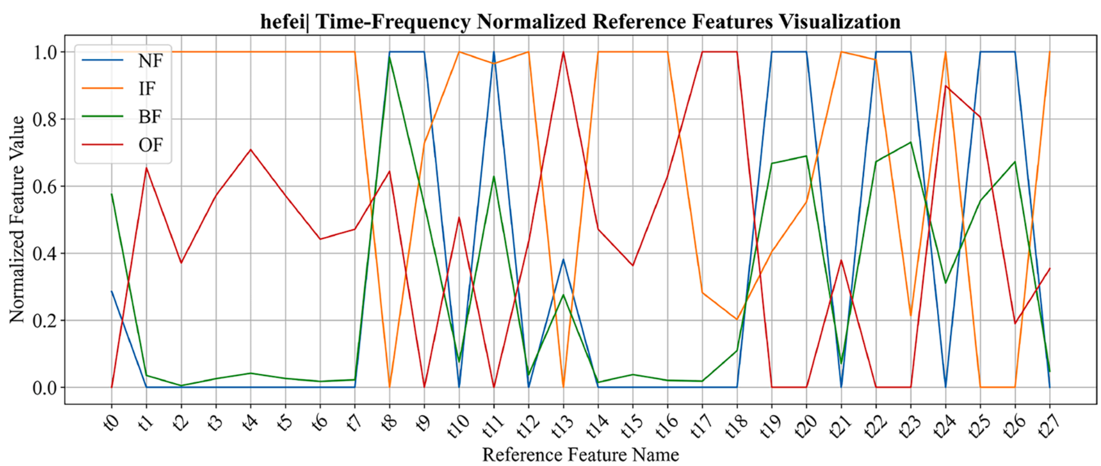
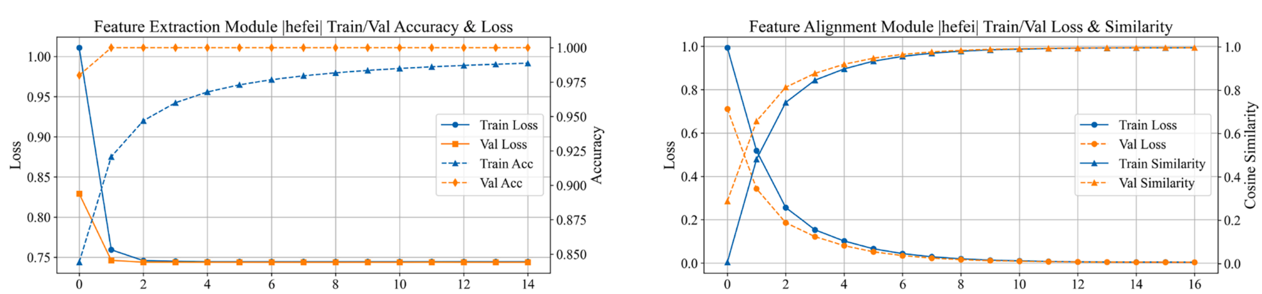


Fig. 4. Visualization of normalized time-frequency reference features..



(a) Extraction Train/Val Values (b) Alignment Train/Val Values

Fig. 5. Training and validation curves of the proposed modules.

Table 9
Experiment on KPI functions.

Dataset	Indicators	S_{acc}	U_{acc}	H_{acc}
HEFEI	Similarity	**99.70%**	**98.80%**	**99.25%**
	Distance	99.50%	88.60%	93.73%
HUST	Similarity	**95.50%**	**93.00%**	**94.23%**
	Distance	88.75%	90.67%	89.70%

In contrast, FPR^{seen} is more sensitive to the choice of thresholding strategy. When the thresholds become more conservative (e.g., using larger percentile p or larger trimming ratio r), FPR^{seen} increases noticeably, implying that more seen samples are incorrectly rejected as unseen. In practice, most thresholding methods introduce at least one hyper-parameter, and manually tuning these parameters can be unreliable, especially when only limited validation data are available and unseen-fault data are inherently absent during training. Our goal is not

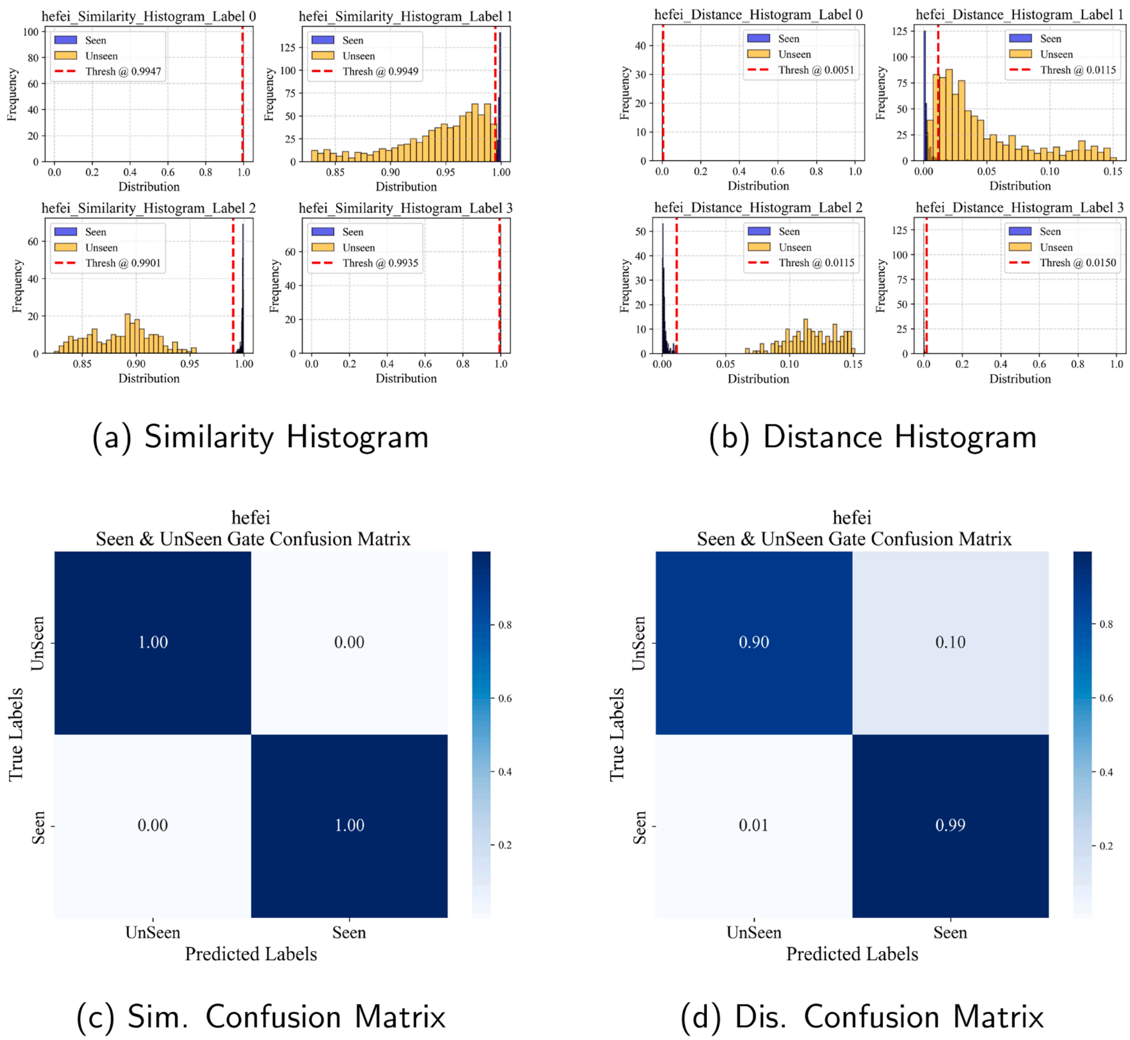


Fig. 6. Histograms and confusion matrices under different KPI functions.

to trade off seen-class recognition accuracy for marginal gains in unseen detection performance.

Moreover, realistic industrial monitoring may involve strong noise and low-quality signals, which could increase score fluctuations near the decision boundary. In such scenarios, incorporating more robust threshold optimization strategies can further enhance the robustness of unseen detection without significantly sacrificing seen-class accuracy.

In addition, the gating decision is implemented via a simple thresholding rule (Score $\geq \tau_S$ indicates a seen sample). As illustrated in Fig. 7, the similarity-optimized scores for seen samples are sharply concentrated near 1, while unseen samples exhibit a broader distribution with lower values. This evident margin demonstrates the discriminative capability of the gating module.

4.2.5. *Unseen database clustering strategy*

To investigate the impact of clustering strategies and hyperparameter settings in the unseen database, comparative experiments were conducted on the HEFEI dataset using DBSCAN and K-means. The results are reported in Table 11. It can be observed that K-means consistently outperforms DBSCAN in terms of overall recognition performance. Although DBSCAN is able to automatically determine the number of clusters, its performance is highly sensitive to additional hyperparameter (e.g., ϵ and $minPts$), making it difficult to tune and leading to unstable clustering quality in this task. In contrast, K-means requires only the cluster number K_c, resulting in a more stable and controllable optimization process. As K_c increases, the recognition performance gradually improves and becomes stable, achieving the best results under an appropriate cluster setting. Considering practical industrial scenarios where multiple unknown fault types rarely occur simultaneously, K_c is typically set slightly larger than the number of seen fault categories to balance clustering flexibility and computational cost.

In the proposed framework, the unseen-fault identification module is triggered periodically when the gated-unseen samples reach a predefined size B_u (i.e., clustering is performed once every B_u gated unseen samples). Therefore, the latency of unseen-fault recognition is closely related to the online sample generation rate determined by the vibration-signal sampling configuration. Assuming the signal sampling frequency is L samples/second, a sliding-window width W, and an overlap ratio R, the number of extracted window samples per second can be estimated as

$$M = \left\lfloor \frac{L - W}{W(1 - R)} \right\rfloor + 1 \tag{30}$$

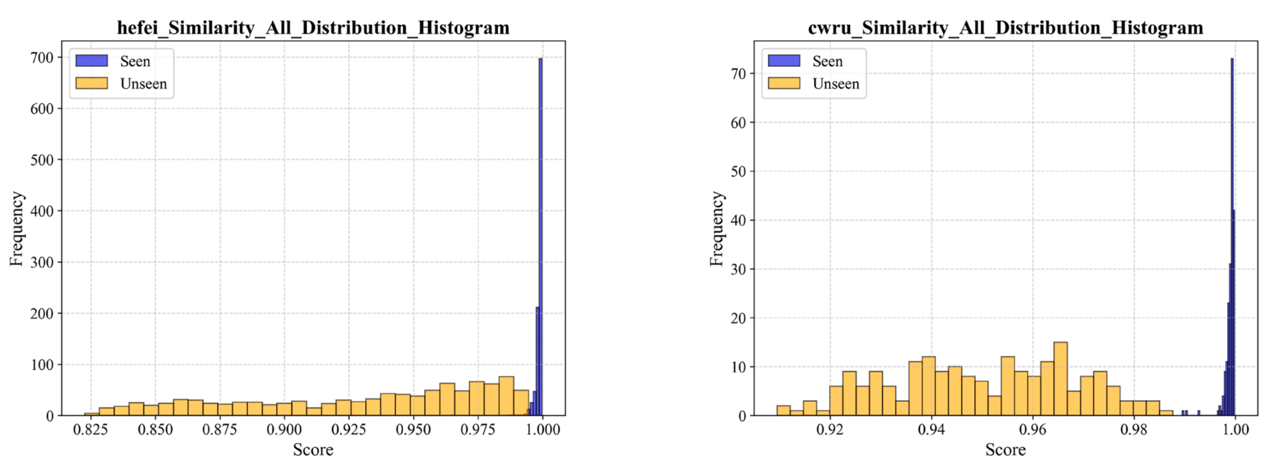
where M denotes the sample generation rate (samples/second). Accordingly, the time required to accumulate B_u gated unseen samples (i.e., the detection delay of the unseen class recognition module) is given by

$$t_{\text{delay}} = \frac{B_u}{M} \tag{31}$$

This indicates that B_u should be chosen with respect to the online sampling rate to avoid backlog and excessive latency. Specifically, set-

Table 10
Parameter experiment (under different thresholding method τ^S).

Dataset	Method	Parameter	τ^n	τ^{if}	τ^{bf}	τ^{of}	FPR^{seen}	TPR^{unseen}	$AUROC^{Gated}$
HEFEI	Proposed	–	**0.996920**	**0.993498**	**0.991448**	**0.99610**	**0.0090**	**0.9980**	**0.9995**
	Percentile	$p = .01$	0.997484	0.993718	0.992160	0.996589	0.0100	0.9980	0.9995
		$p = .05$	0.997850	0.995184	0.995013	0.997936	0.0360	1.0000	0.9995
		$p = .10$	0.998605	0.996026	0.997000	0.998302	0.0890	1.0000	0.9995
	Trimmed-Min	$r = 0.01$	0.996920	0.993498	0.991448	0.996104	0.0090	0.9980	0.9995
		$r = 0.05$	0.997753	0.994785	0.994755	0.997713	0.0300	1.0000	0.9995
		$r = 0.10$	0.998580	0.996009	0.996951	0.998284	0.0860	1.0000	0.9995
	EVT-based	$r = 0.05,\ \alpha = 0.01$	0.997035	0.993573	0.991587	0.996132	0.0090	0.9980	0.9995
		$r = 0.10,\ \alpha = 0.05$	0.997198	0.99366	0.991773	0.996623	0.0110	0.9980	0.9995
		$r = 0.15,\ \alpha = 0.10$	0.997329	0.993771	0.991984	0.996914	0.0140	0.9980	0.9995
	FPR-based	$p = .01$	0.997484	0.993718	0.992160	0.996589	0.0100	0.9980	0.9995
		$p = .05$	0.9995	0.995184	0.995013	0.997936	0.0360	1.0000	0.9995
		$p = .10$	0.998605	0.996026	0.997000	0.998302	0.0890	1.0000	0.9995
CWRU	Proposed	–	**0.996361**	**0.996600**	**0.989323**	**0.992244**	**0.0050**	**1.0000**	**1.0000**
	Percentile	$p = .01$	0.996460	0.996600	0.989323	0.992244	0.0100	1.0000	1.0000
		$p = .05$	0.997297	0.997785	0.995693	0.996739	0.0200	1.0000	1.0000
		$p = .10$	0.997753	0.998412	0.997895	0.998200	0.0800	1.0000	1.0000
	Trimmed-Min	$r = 0.01$	0.996361	0.996600	0.989323	0.992244	0.0050	1.0000	1.0000
		$r = 0.05$	0.997076	0.997472	0.994535	0.99606	0.0200	1.0000	1.0000
		$r = 0.10$	0.997643	0.998395	0.997866	0.997916	0.0650	1.0000	1.0000
	EVT-based	$r = 0.05,\ \alpha = 0.01$	0.996396	0.996645	0.9893207	0.992291	0.0050	1.0000	1.0000
		$r = 0.10,\ \alpha = 0.05$	0.996444	0.996705	0.990651	0.992388	0.0100	1.0000	1.0000
		$r = 0.15,\ \alpha = 0.10$	0.9964978	0.996774	0.991576	0.992522	0.0100	1.0000	1.0000
	FPR-based	$p = .01$	0.996460	0.996827	0.991575	0.992254	0.0100	1.0000	1.0000
		$p = .05$	0.997297	0.997785	0.995693	0.996739	0.0200	1.0000	1.0000
		$p = .10$	0.997753	0.998412	0.997895	0.998200	0.0800	1.0000	1.0000



(a) HEFEI Score Distribution (b) CWRU Score Distribution

Fig. 7. Score distribution of seen and unseen samples under similarity-based gating on HEFEI and CWRU datasets.

Table 11
Parameter experiment (unseen database clustering).

Dataset	Method	Evaluation Metrics	$minPts$	$\epsilon = 0.1$	$\epsilon = 0.2$	$\epsilon = 0.3$	$\epsilon = 0.4$	$\epsilon = 0.5$
HEFEI		S_{acc}	–	**99.80%**	**99.80%**	**99.80%**	**99.80%**	**99.80%**
	DBSCAN	U_{acc} | H_{acc}	3	27.90% | 43.61%	77.00% | 86.93%	73.60% | 84.72%	73.90% | 84.92%	50.00% | 66.64%
		U_{acc} | H_{acc}	4	25.00% | 39.98%	77.30% | 87.12%	73.60% | 84.72%	73.90% | 84.92%	50.00% | 66.64%
		U_{acc} | H_{acc}	5	25.00% | 39.98%	**78.20% | 87.69%**	73.90% | 84.92%	73.90% | 84.92%	50.00% | 66.64%
		U_{acc} | H_{acc}	6	25.00% | 39.98%	77.40% | 87.18%	73.90% | 84.92%	73.90% | 84.92%	50.00% | 66.64%
		U_{acc} | H_{acc}	7	25.00% | 39.98%	76.50% | 86.61%	73.70% | 84.79%	73.90% | 84.92%	50.00% | 66.64%
			$K_c = 1$	$K_c = 2$	$K_c = 3$	$K_c = 4$	$K_c = 5$	$K_c = 6$
	KMeans	U_{acc} | H_{acc}	25.00% | 39.98%	50.00% | 66.62%	73.70% | 84.79%	**97.20% | 98.48%**	**97.20% | 98.48%**	**97.20% | 98.48%**

ting a relatively small B_u (compared with the number of samples collected per second) ensures that the unseen-fault identification can be executed in near real-time while maintaining bounded memory usage and stable throughput. In our implementation, the runtime of clustering and identification is required to satisfy $t_{\text{proc}}(B_u) < t_{\text{delay}}$, which guarantees that the system can process the incoming gated-unseen stream without accumulating delay.

However, an overly small B_u may degrade clustering quality, since the unseen clusters are estimated from insufficient samples and are therefore more sensitive to noise and sample variability. This may lead to unstable cluster prototypes and reduced unseen-fault identification accuracy. Consequently, the choice of B_u should strike a balance between real-time responsiveness and clustering reliability, and it is typically set to a moderate value that guarantees both acceptable detection delay and stable cluster formation.

For example, under $L = 51{,}200$ samples/second, $W = 2048$, and $R = 0.8$, the proposed sliding-window extraction produces approximately $M \approx 121$ samples/second. Thus, setting $B_u = 50$ results in a triggering interval of $t_{\text{delay}} \approx 0.4132$ s, while $B_u = 100$ leads to $t_{\text{delay}} \approx 0.8264$ s. This provides an explicit trade-off between recognition stability (larger B_u) and online latency (smaller B_u).

In practice, B_u is selected according to the following principles:

- Real-time constraint: $t_{\text{proc}}(B_u) < t_{\text{delay}}$ to avoid backlog accumulation.
- Clustering stability: B_u should be sufficiently large to guarantee stable cluster formation.

Under our setting, $B_u \in [50, 200]$ provides a good trade-off between latency and clustering reliability.

4.2.6. Computational complexity and inference efficiency

To evaluate the practicality of the proposed framework for industrial deployment, we conduct a computational complexity and inference-efficiency analysis on both the HEFEI and CWRU datasets. The results are reported in Table 12. Experiments are performed on an Apple M2 Pro-platform, where the average end-to-end inference latency per sample is approximately 120–140 ms, indicating that the proposed framework can satisfy real-time deployment requirements.

In addition, we investigate the influence of delayed identification in the unseen database. Specifically, the unseen-fault identification procedure is triggered once every B_u gated unseen samples are accumulated (i.e., the system performs one identification step when the number of collected unseen samples reaches a multiple of B_u). Since the evaluation datasets are relatively small, no pruning strategy is applied in this experiment. In practical industrial scenarios, however, the unseen database may continuously grow, making it necessary to adopt pruning or online updating strategies to avoid excessive computational overhead. We evaluate four settings ($B_u \in \{50, 100, 150, 200\}$). The results show that increasing B_u leads to slightly higher memory usage and identification latency, while the overall runtime remains comparable across different settings.

Furthermore, the computational complexity of the proposed framework is analyzed. Since the computational overhead of other modules is negligible compared with the feature extraction backbone, the overall FLOPs are dominated by the feature extraction network and amount to approximately 1.05×10^9 FLOPs per test sample. The FLOPs of a convolutional layer are computed as:

$$\text{FLOPs} = 2 \times C_{\text{in}} \times C_{\text{out}} \times H_{\text{out}} \times W_{\text{out}} \times K^2 \tag{32}$$

4.2.7. Noise robustness experiment

To evaluate the robustness and stability of the proposed model under noisy conditions, Gaussian white noise with different signal-to-noise ratios (SNRs) of 5 dB, 10 dB, 15 dB, 20 dB, 25 dB, and 30 dB was added to the test data for comparative experiments. The experimental results are reported in Table 13.

As can be observed, the proposed model maintains consistently high recognition accuracy across all noise levels, demonstrating strong robustness against Gaussian noise. Notably, the model achieves its highest recognition performance at an SNR of 30 dB.

4.2.8. Unseen faults experiment

Figs. 8 and 9 show the overall recognition performance of TransZeroNet on the HEFEI and HUST datasets, achieving H_{acc} scores of 99.25% and 94.23%, respectively.

4.2.9. Variable operating experiment

To evaluate the model's generalization in zero-shot fault diagnosis under variable operating conditions, a CWRU dataset with 40 condition combinations was constructed in Table 14.

Seen classes are constructed from a randomly selected sampling frequency, using samples with the same horsepower but different fault diameters and types, along with the normal condition. Unseen classes are built from another frequency, prioritizing faults matching the seen classes in type and diameter to enhance distribution discrepancy. Remaining unseen classes are randomly added to reach a total of N_u classes ($0 < N_u < 33$). Detailed settings are shown in Table 6.

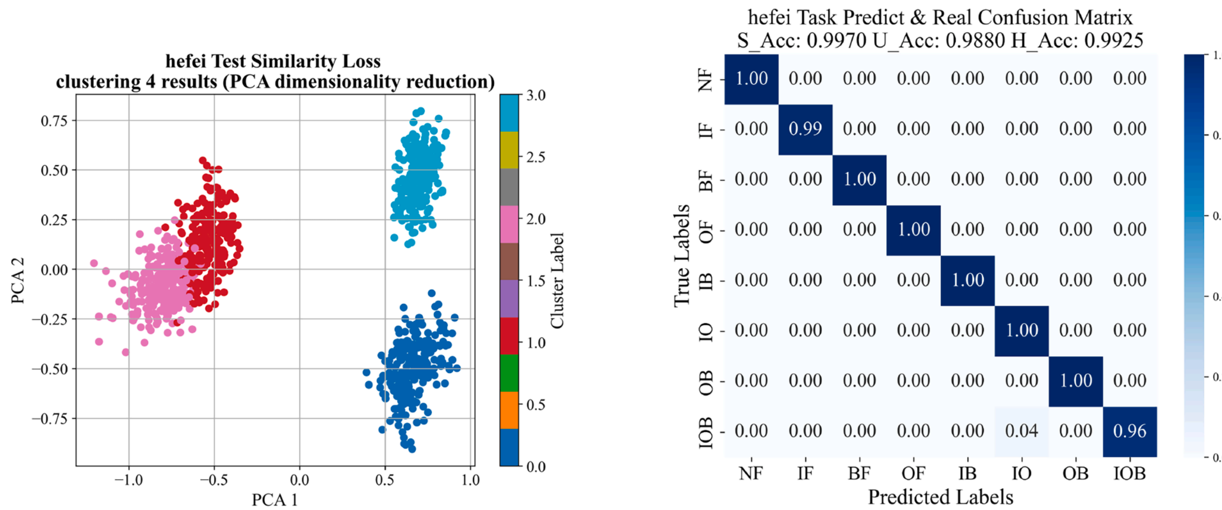
To investigate the impact of the number of unseen classes on the performance of the proposed model, N_u was varied as $1, 2, 3, 4, 5, 6, 7$ with all other settings fixed. The results are reported in Table 15.

As the number of unseen classes N_u increases, the proposed model consistently achieves a 100% fault detection rate, demonstrating stable fault identification. Although the harmonic mean accuracy (H_{acc}) slightly decreases, it remains above 88.83% across all settings, indicating robust performance despite increased task complexity. These results validate the model's strong generalization and practical effectiveness for GZSL fault diagnosis.

In addition, unseen classes were constructed under the same sampling frequency as seen classes to evaluate the framework under more challenging distribution overlap conditions. The corresponding gating thresholds and performance are reported in Table 16. Since the seen and unseen distributions become more similar, the separation margin is reduced, which weakens the gating discriminability and leads to a noticeable drop in overall performance, particularly on unseen faults. Nevertheless, the proposed gating module still demonstrates reliable capability in detecting abnormal samples. This suggests that the method

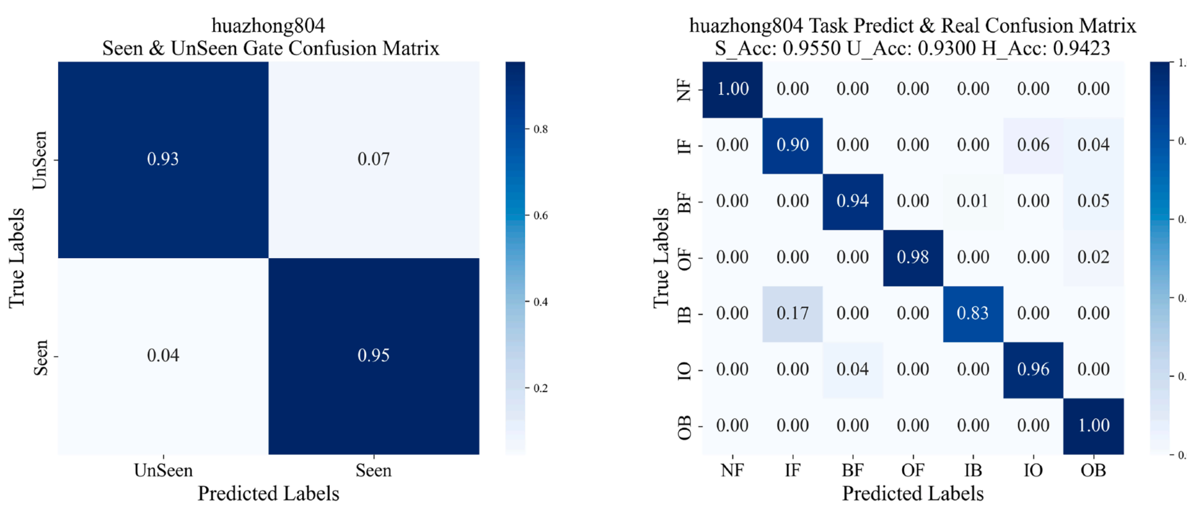
Table 12
Computational complexity, runtime, and memory analysis.

<table>
<tr><th>Dataset</th><th>Modes</th><th>Kinds</th><th>Samples</th><th>Time (All/S)</th><th>Time (Avg/MS)</th><th>Memory (All/MB)</th><th>$\text{FLOPs}_{\text{test}}$ (Avg/FLOPs)</th><th>N_u (Samples)</th><th>$\text{Time}_{\text{unseen}}$ (MS)</th><th>$\text{Memory}_{\text{unseen}}$ (Once/MB)</th><th>Time (All/S)</th><th>Time (Avg/MS)</th><th>Memory (All/MB)</th></tr>
<tr><td rowspan="4">HEFEI</td><td rowspan="2">Train</td><td rowspan="2">4</td><td rowspan="2">298 | 75</td><td rowspan="2">1065.35</td><td rowspan="2">714.04</td><td rowspan="2">238.26</td><td rowspan="2">–</td><td>50</td><td>102.44→928.83</td><td>$2.97_{\max}$</td><td>286.79</td><td>143.40</td><td>494.61</td></tr>
<tr><td>100</td><td>195.35→930.93</td><td>$7.90_{\max}$</td><td>287.49</td><td>143.75</td><td>587.53</td></tr>
<tr><td rowspan="2">Test</td><td rowspan="2">8</td><td rowspan="2">250</td><td rowspan="2">279.51</td><td rowspan="2">139.75</td><td rowspan="2">459.25</td><td rowspan="2">≈ 1.05e9</td><td>150</td><td>280.59→836.73</td><td>$7.65_{\max}$</td><td>280.32</td><td>140.16</td><td>612.46</td></tr>
<tr><td>200</td><td>407.94→960.30</td><td>$7.93_{\max}$</td><td>281.45</td><td>140.72</td><td>521.66</td></tr>
<tr><td rowspan="4">CWRU</td><td rowspan="2">Train</td><td rowspan="2">4</td><td rowspan="2">195 | 49</td><td rowspan="2">1050.32</td><td rowspan="2">484.46</td><td rowspan="2">107.27</td><td rowspan="2">–</td><td>50</td><td>103.93→147.84</td><td>$0.15_{\max}$</td><td>52.98</td><td>132.44</td><td>683.55</td></tr>
<tr><td>100</td><td>326.45</td><td>0.42</td><td>52.62</td><td>131.55</td><td>758.18</td></tr>
<tr><td rowspan="2">Test</td><td rowspan="2">8</td><td rowspan="2">50</td><td rowspan="2">48.31</td><td rowspan="2">120.77</td><td rowspan="2">716.12</td><td rowspan="2">≈ 1.05e9</td><td>150</td><td>347.38</td><td>6.63</td><td>52.88</td><td>132.21</td><td>796.87</td></tr>
<tr><td>–</td><td>–</td><td>–</td><td>–</td><td>–</td><td>–</td></tr>
</table>



(a) Clustering Result (b) HEFEI Result

Fig. 8. Final results of the unseen faults experiment on HEFEI Dataset.



(a) Gated Detection Result (b) HUST Result

Fig. 9. Final results of the unseen faults experiment on HUST Dataset.

Table 13
H_{acc} Under different noise levels.

Dataset	5 dB	10 dB	15 dB	20 dB	25 dB	30 dB
HEFEI	98.60%	99.10%	99.00%	98.95%	99.20%	**99.25%**

remains effective, but its performance is constrained when the seen/unseen distributions are highly overlapped.

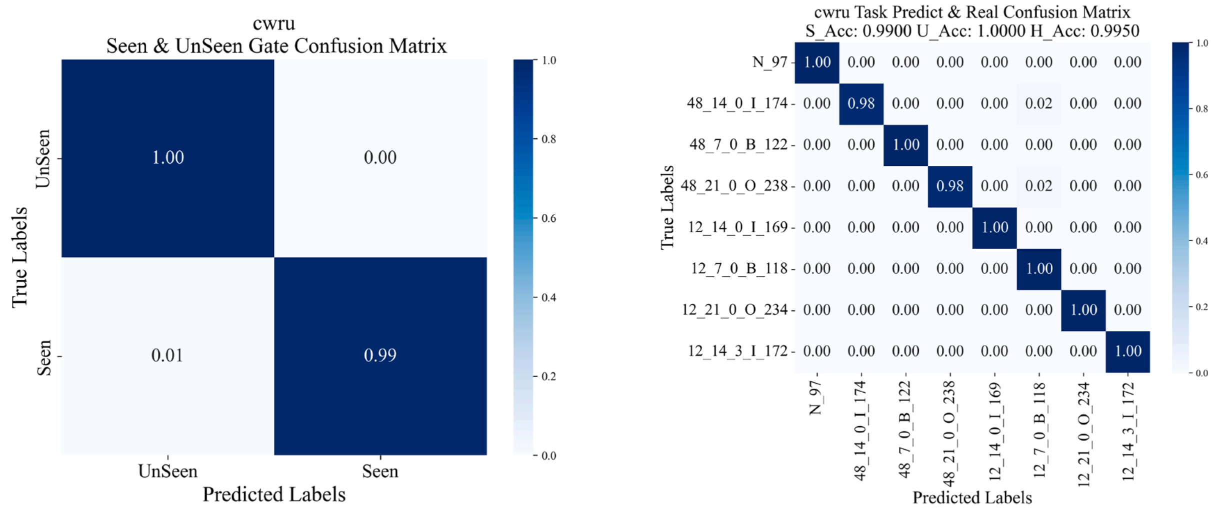
To fairly compare with previous zero-shot fault experiments, the number of unseen fault categories N_u was set to 4. Fig. 10b shows the overall recognition performance of TransZeroNet on the CWRU dataset, achieving a harmonic mean accuracy (H_{acc}) of 99.50%.

4.2.10. Comparative study result and analysis

To further validate the effectiveness of the proposed TransZeroNet, comparative experiments are conducted on the HEFEI dataset, and the results are summarized in Table 17.

(A) GZSL baselines. Comparative experiments are conducted with representative generative ZSL/GZSL methods, including CADA-VAE (Schonfeld et al., 2019), CE-GZSL (Han et al., 2021), and FREE (Chen et al., 2021b). For a fair comparison, the semantic descriptors of all methods are constructed using the same normalized time-frequency reference features. However, under our few-shot training setting, conventional generative embedding-based GZSL methods often fail to learn stable semantic alignment between seen and unseen fault classes, and the generated pseudo samples may suffer from semantic shift relative to real fault data. In contrast, our method leverages a dedicated gating mechanism to reliably separate seen and unseen samples, enabling unseen-fault inference on real test data and leads to more balanced GZSL performance.

(B) OSR / OOD baselines (Class-Gated Recognition Module). In addition, the proposed class-gated module is further compared with representative open-set recognition and out-of-distribution baselines, including MSP (maximum softmax probability) (Hendrycks & Gimpel,



(a) Gated Detection Result (b) CWRU Result

Fig. 10. Final results under variable operating conditions.

Table 14
CWRU database.

Sampling Speed	Fault Diameter	Motor Load (HP)	Health Status
12K	0.007	0	IF,BF,OF
	0.007	3	IF,BF,OF
	0.014	0	IF,BF,OF
	0.014	3	IF,BF,OF
	0.021	0	IF,BF,OF
	0.021	3	IF,BF,OF
48K	0.007	0	IF,BF,OF
	0.007	3	IF,BF,OF
	0.014	0	IF,BF,OF
	0.014	3	IF,BF,OF
	0.021	0	IF,BF,OF
	0.021	3	IF,BF,OF
-	–	0	H
	–	1	H
	–	2	H
	–	3	H

Table 15
Unseen class quantity experiment.

N_u	Normal	Fault	S_{acc}	U_{acc}	H_{acc}
1	98.00%	100.00%	99.00%	100.00%	99.50%
2	98.00%	100.00%	98.50%	100.00%	99.24%
3	98.00%	100.00%	99.00%	100.00%	99.50%
4	100.00%	100.00%	99.00%	100.00%	99.50%
5	100.00%	100.00%	99.00%	100.00%	99.50%
6	100.00%	100.00%	100.00%	98.67%	99.33%
7	98.00%	100.00%	96.50%	82.29%	88.83%

2017), Energy-based (Liu et al., 2020), ARPL (Chen et al., 2021a), and OpenMax (LibMR/EVT) (Scheirer et al., 2011). OSR/OOD baselines mainly rely on confidence-based rejection or calibrated score modeling. However, MSP suffers from over-confident predictions on seen classes, causing score saturation and a limited thresholding range, which may retain many unseen samples as seen. In contrast, our gate measures distribution discrepancy in the representation space, making it less sensitive to confidence saturation and more discriminative for seen/unseen separation, achieving the best $AUROC^{gated}$ with high TPR^{unseen} and low FPR^{seen}.

4.2.11. *Ablation and sensitivity study*

To isolate the unique contribution of TransZeroNet and better understand its key design choices, sensitivity experiments and ablation studies are conducted on the reference representations and the Feature Alignment Module, respectively.

First, Table 18 reports the results when replacing the seen-class reference vectors with randomly generated vectors of different dimensions, as well as the TF-based reference vectors. The performance remains consistently high across different settings, indicating that the proposed framework is not strictly dependent on high-quality semantic attributes or a specific reference design. Instead, the seen-class reference vectors mainly serve as a distribution benchmark, and the gate focuses on quantifying the deviation of test samples from this benchmark.

To further validate which component is essential for enabling such deviation-based discrimination, Table 19 provides an ablation study on the Feature Alignment Module. When removing the alignment module (A1), the gated separation becomes ineffective, resulting in severe performance degradation in both gating metrics (FPR^{seen}, $AUROC^{Gated}$) and GZSL recognition performance (U_{acc} and H_{acc}). This confirms that although the reference vectors can be flexible, the proposed alignment mechanism is the critical component that enforces reference consistency and enables reliable seen/unseen discrimination.

These results jointly suggest that TransZeroNet does not rely on semantics in the conventional ZSL sense. Instead, its generalization ability mainly stems from learning a reference-aligned representation space, in which the similarity-based KPI becomes highly separable for seen and unseen samples. The sensitivity study demonstrates that the reference vector acts as a relative anchor rather than a strict semantic descriptor, while the ablation results verify that the Feature Alignment Module is indispensable to achieve stable deviation quantification. Without such alignment, the learned features cannot be consistently projected toward class references, leading to an unreliable KPI distribution and thus failure in open-set gating and unseen-fault identification.

Table 16
Results on the CWRU dataset at 12 kHz.

Dataset	$\text{Seen}_{\text{Sample}}$ (Name)	$\text{Unseen}_{\text{Sample}}$ (Name)	$\text{Gated}_{\text{normal}}$	$\text{Gated}_{\text{fault}}$	S_{Acc}	U_{Acc}	H_{Acc}
CWRU	N_97 12_7_0_I_105	12_7_3_I_108 12_21_3_B_225	100.00%	100.00%	99.00%	77.00%	**86.62%**
	12_21_0_B_222 12_14_0_O_197	12_14_3_O_200 12_7_0_O_130	Threshold→	τ^{n} 0.997661	τ^{if} 0.997318	τ^{bf} 0.970993	τ^{of} 0.991292
	N_100 12_21_0_I_209	12_21_3_I_212 12_7_3_B_121	100.00%	98.00%	91.50%	86.50%	**88.93%**
	12_7_0_B_118 12_14_0_O_197	12_14_3_O_200 12_7_0_I_105	Threshold→	τ^{n} 0.994092	τ^{if} 0.998369	τ^{bf} 0.997787	τ^{of} 0.997791

Table 17
Comparison of Model Performance on HEFEI Dataset.

No.	Experimental method	Metric-1	Metric-2	Metric-3
(A) GZSL baselines				
		S_{acc}	U_{acc}	H_{acc}
1	CADA-VAE	47.40%	22.00%	30.05%
2	CE-GZSL	99.60%	39.50%	56.57%
3	FREE	**99.98%**	31.20%	47.56%
4	**TransZeroNet (Ours)**	99.70%	**98.80%**	**99.25%**
(B) OSD / OOD baselines (Class-Gated Recognition Module)				
		FPR^{seen}	TPR^{unseen}	$AUROC^{gated}$
5	MSP	**0.0030**	0.3670	0.6821
6	Energy-based score (EBO)	0.0500	**1.0000**	0.9989
7	ARPL	0.0240	0.9470	0.9929
8	OpenMax (LibMR/EVT)	0.0179	0.8299	0.9727
9	**TransZeroNet (Ours)**	0.0090	0.9980	**0.9995**

Table 18
Sensitivity study on reference representations.

Dataset	Reference	Evaluation Metrics	$Dim = 5$	$Dim = 10$	$Dim = 15$	$Dim = 20$	$Dim = 25$	$Dim = 28$
HEFEI	Random	$\text{Gated}_{\text{seen}}$	100.00%	99.00%	**100.00%**	99.00%	100.00%	100.00%
		$\text{Gated}_{\text{unseen}}$	99.00%	98.00%	**100.00%**	100.00%	99.00%	100.00%
		H_{acc}	99.00%	98.34%	**99.25%**	98.85%	85.09%	99.15%
	TF	$\text{Gated}_{\text{seen}}$	99.00%	100.00%	100.00%	100.00%	99.00%	**100.00%**
		$\text{Gated}_{\text{unseen}}$	91.00%	100.00%	99.00%	100.00%	99.00%	**100.00%**
		H_{acc}	94.31%	99.20%	98.79%	98.54%	98.74%	**99.25%**

Table 19
Ablation study of key components in TransZeroNet.

ID	Variant	Alignment	FPR^{seen}	TPR^{unseen}	$AUROC^{Gated}$	S_{acc}	U_{acc}	H_{acc}
A0	Full (TransZeroNet)	✓	0.0040	0.9780	0.9993	99.60%	96.70%	98.13%
A1	w/o Feature Alignment	✗	1.0000	1.0000	0.5934	74.90%	25.00%	37.49%

5. Conclusion

To address the limitations of existing GZSL fault diagnosis methods, such as heavy semantic dependency, limited unseen-fault flexibility, and high training cost, this paper proposes TransZeroNet, a unified framework for generalized zero-shot fault diagnosis under both unseen-fault and variable operating-condition scenarios. TransZeroNet introduces a distribution-discrepancy metric and a class-gated recognition module to quantify the deviation between test samples and learned class reference features, thereby alleviating classification bias between seen and unseen data without requiring high-quality manual semantics.

In addition, an Unseen Database is designed to accumulate detected unseen samples, and an unsupervised inference module is further adopted for fine-grained unseen fault identification, avoiding the domain shift issues commonly observed in generative ZSL methods. Extensive experiments on HEFEI, HUST, and CWRU demonstrate the effectiveness and robustness of the proposed framework, achieving accuracies of 99.25%, 94.23%, and 99.50%, respectively.

CRediT authorship contribution statement

Jing Wang: Conceptualization, Methodology, Investigation, Validation, Formal analysis, Project administration, Funding acquisition, Writing – original draft, Writing – review & editing; **Ning Li:** Conceptualization, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing; **Meng Zhou:** Conceptualization, Resources, Supervision, Project administration; **Rong Su:** Conceptualization, Supervision, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant 62373005 and 62273007, and in part by the National Research Foundation Singapore under its AI Singapore Programme, Award AISG2-GC-2023-007.

References

Bai, X., Ma, Z., & Meng, G. (2022). Bearing fault diagnosis based on wavelet transform and residual shrinkage network. In *2022 International conference on computer network, electronic and automation (ICCNEA)* (pp. 373–378). https://doi.org/10.1109/ICCNEA57056.2022.00087.

Basu, S., Campbell, R. H., & Karahalios, K. (2023). Detection of novel COVID-19 variants with zero-shot learning. In *2023 IEEE 11th international conference on healthcare informatics (ICHI)* (pp. 597–600). https://doi.org/10.1109/ICHI57859.2023.00106.

Cai, L., Yin, H., Lin, J., Zhao, D., & Qin, Y. (2024). A multiattribute learning model for zero-sample mechanical fault diagnosis. *IEEE Transactions on Industrial Informatics*, *20*(7), 9633–9643. https://doi.org/10.1109/TII.2024.3383459.

Cen, J., Zhao, B., Liu, X., Li, X., Deng, F., & Huang, H. (2025). Generalized zero-shot learning based on diffusion model and multilabel network for compound fault diagnosis. *IEEE Transactions on Industrial Informatics*, (pp. 1–12). https://doi.org/10.1109/TII.2025.3567388.

Chen, G., Peng, P., Wang, X., & Tian, Y. (2021a). Adversarial reciprocal points learning for open set recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, https://doi.org/10.1109/TPAMI.2021.3106743.

Chen, H., Hong, J., Cheng, C., Kong, Y., & Zheng, X. (2024a). Zero-shot compound fault diagnosis method using semantic construction and embedding. *IEEE Transactions on Instrumentation and Measurement*, *73*, 1–13. https://doi.org/10.1109/TIM.2024.3470961.

Chen, J., & Yang, R. (2024). A semantic-consistent conditional generative adversarial model for zero-shot learning in bearing fault diagnosis. In *2024 IEEE 13th data driven control and learning systems conference (DDCLS)* (pp. 1044–1049). https://doi.org/10.1109/DDCLS61622.2024.10606578.

Chen, S., Wang, W., Xia, B., Peng, Q., You, X., Zheng, F., & Shao, L. (2021b). Free: Feature refinement for generalized zero-shot learning. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)* (pp. 122–131).

Chen, Y., Tao, L., Liu, X., Ma, J., Lu, C., & Liu, H. (2024b). Open-set fault recognition and inference for rolling bearing based on open fault semantic subspace. *IEEE Transactions on Instrumentation and Measurement*, *73*, 1–11. https://doi.org/10.1109/TIM.2023.3336436.

Chen, Z., Wu, J., Deng, C., Wang, X., & Wang, Y. (2023). Deep attention relation network: A zero-shot learning method for bearing fault diagnosis under unknown domains. *IEEE Transactions on Reliability*, *72*(1), 79–89. https://doi.org/10.1109/TR.2022.3177930.

Fang, L., Liu, Y., Li, X., & Chang, J. (2024). Intelligent fault diagnosis of rolling bearing based on deep transfer learning. In *2024 6th international conference on natural language processing (ICNLP)* (pp. 753–757). https://doi.org/10.1109/ICNLP60986.2024.10692521.

Han, Z., Fu, Z., Chen, S., & Yang, J. (2021). Contrastive embedding for generalized zero-shot learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 2371–2381).

Hendriks, J., Dumond, P., & Knox, D. A. (2022). Towards better benchmarking using the CWRU bearing fault dataset. *Mechanical Systems and Signal Processing*, *169*, 108732.

Hendrycks, D., & Gimpel, K. (2017). A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *5th international conference on learning representations, ICLR 2017, toulon, france, april 24–26, 2017, conference track proceedings*. OpenReview.net. https://openreview.net/forum?id=Hkg4TI9xl.

Huang, J., Li, Z., Ye, L., & Zhou, Z. (2021). Fault classification of industrial processes based on generalized zero-shot learning. In *2021 IEEE 10th data driven control and learning systems conference (DDCLS)* (pp. 887–892). https://doi.org/10.1109/DDCLS52934.2021.9455689.

Huang, J., Ma, L., & Wang, J. (2023). Unknown fault diagnosis based on transfer learning under multiple working conditions. In *2023 China automation congress (CAC)* (pp. 6489–6494). https://doi.org/10.1109/CAC59555.2023.10450456.

Kömürcü, K., & Petkevičius, L. (2024). Zero shot classification for change detection in satellite imagery. In *2024 IEEE 11th workshop on advances in information, electronic and electrical engineering (AIEEE)* (pp. 1–6). https://doi.org/10.1109/AIEEE62837.2024.10586705.

Lee, J.-C., Shibu, A., & Lee, D.-G. (2023). Adaptive margin-based contrastive network for generalized zero-shot learning. In *2023 IEEE International conference on consumer electronics (ICCE)* (pp. 1–4). https://doi.org/10.1109/ICCE56470.2023.10043553.

Liu, W., Li, Y., Ma, R., Qi, Y., & Liu, Y. (2024). A generative model for zero-shot learnning fault diagnosis of rolling bearings. In *2024 IEEE 6th advanced information management, communicates, electronic and automation control conference (IMCEC)* (pp. 1214–1219). (*vol. 6*). https://doi.org/10.1109/IMCEC59810.2024.10575185.

Liu, W., Wang, X., Owens, J., & Li, Y. (2020). Energy-based out-of-distribution detection. In H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, & H. Lin (Eds.), *Advances in neural information processing systems* (pp. 21464–21475). Curran Associates, Inc. (*vol. 33*). https://proceedings.neurips.cc/paper_files/paper/2020/file/f5496252609c43eb8a3d147ab9b9c006-Paper.pdf.

Liu, Z., Yang, R., Chen, X., & Xue, Y. (2022). Weighted multi-view zero-shot learning prototype shift model in fault diagnosis. In *2022 China automation congress (CAC)* (pp. 5322–5327). https://doi.org/10.1109/CAC57257.2022.10055280.

Long, J., Lin, J., Jiang, L., Yang, Z., Guo, J., Yin, T., & Li, C. (2024). Masked autoencoder via end-to-end zero-shot learning for fault diagnosis of unseen classes. *IEEE Transactions on Instrumentation and Measurement*, *73*, 1–10. https://doi.org/10.1109/TIM.2024.3472905.

Luo, M., Xu, J., Fan, Y., & Zhang, J. (2023). Trnet: A cross-component few-shot mechanical fault diagnosis. *IEEE Transactions on Industrial Informatics*, *19*(5), 6883–6894. https://doi.org/10.1109/TII.2022.3204554.

Ma, L., Shi, F., Wu, Z., & Peng, K. (2023). A survey of few-shot learning-based compound fault diagnosis methods for industrial processes. In *2023 IEEE 6th international conference on industrial cyber-physical systems (ICPS)* (pp. 1–4). https://doi.org/10.1109/ICPS58381.2023.10128105.

Pan, P., Li, Y., & Zhao, D. (2022). Generalized zero-shot learning fault diagnosis framework based on anomaly detection and contractive stacked autoencoder. In *2022 China automation congress (CAC)* (pp. 2427–2432). https://doi.org/10.1109/CAC57257.2022.10055111.

Qin, Y., Wang, L., Qian, Q., & Mao, Y. (2024). Zero-shot attribute consistent model for bearing fault diagnosis under unknown domain. *IEEE Transactions on Instrumentation and Measurement*, *73*, 1–11. https://doi.org/10.1109/TIM.2024.3378256.

Scheirer, W. J., Rocha, A., Michaels, R., & Boult, T. E. (2011). Meta-recognition: The theory and practice of recognition score analysis. *IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)*, *33*, 1689–1695.

Schonfeld, E., Ebrahimi, S., Sinha, S., Darrell, T., & Akata, Z. (2019). Generalized zero- and few-shot learning via aligned variational autoencoders. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 8247–8255).

Tang, H., Jing, W., Tang, D., Yang, Z., Yang, X., & Xie, W. (2025). Global-local attention-aware zero-shot learning for industrial fault diagnosis. *IEEE Transactions on Instrumentation and Measurement*, *74*, 1–16. https://doi.org/10.1109/TIM.2025.3550233.

Thuan, N. D., & Hong, H. S. (2023). Hust bearing: A practical dataset for ball bearing fault diagnosis. *BMC Research Notes*, *16*(1), 138.

Wang, J., Xu, Z., Niu, F., Liu, J., & Zhu, Z. (2024). Adaptively weighted semantic autoencoder for zero-shot compound fault diagnosis. *IEEE Sensors Journal*, *24*(22), 37472–37481. https://doi.org/10.1109/JSEN.2024.3470515.

Wang, K., Gou, Y., Chen, Z., Wang, J., & Deng, Y. (2023). Complex fault diagnosis of rolling bearings based on improved residual shrinkage networks. In *2023 3rd international conference on robotics, automation and intelligent control (ICRAIC)* (pp. 344–349). https://doi.org/10.1109/ICRAIC61978.2023.00066.

Wu, Q., Ma, Y., Feng, Z., Yang, S., & Hu, H. (2025). Unsupervised contrastive learning-based single domain generalization method for intelligent bearing fault diagnosis. *IEEE Sensors Journal*, *25*(2), 3923–3934. https://doi.org/10.1109/JSEN.2024.3507817.

Xu, J., Kong, H., Li, K., & Ding, X. (2024a). Generative zero-shot compound fault diagnosis based on semantic alignment. *IEEE Transactions on Instrumentation and Measurement*, *73*, 1–13. https://doi.org/10.1109/TIM.2023.3345912.

Xu, J., & Li, K. (2021). Generative zero-shot learning compound fault diagnosis of bearings. In *2021 International conference on sensing, measurement & data analytics in the era of artificial intelligence (ICSMD)* (pp. 1–7). https://doi.org/10.1109/ICSMD53520.2021.9670760.

Xu, J., Zhang, H., Zhou, L., & Fan, Y. (2023). Zero-shot compound fault diagnosis method based on semantic learning and discriminative features. *IEEE Transactions on Instrumentation and Measurement*, *72*, 1–13. https://doi.org/10.1109/TIM.2023.3280503.

Xu, Z., Ma, Y., Pan, Z., & Zheng, X. (2024b). Deep spiking residual shrinkage network for bearing fault diagnosis. *IEEE Transactions on Cybernetics*, *54*(3), 1608–1613. https://doi.org/10.1109/TCYB.2022.3227363.

Yang, X., Guo, T., Huang, J., Zhao, Y., Zhou, X., & Peng, K. (2025). Improved wavelet zero-shot learning model with signal-sensitive features and weighted semantics for compound fault diagnosis of rolling bearing. *IEEE Transactions on Instrumentation and Measurement*, *74*, 1–10. https://doi.org/10.1109/TIM.2025.3548779.

Zhang, J. Q., & Yan, Y. (2006). Detecting the blockage of the sensing lines of a differential-pressure flow sensor in a dynamic process using wavelet transform techniques. *IEEE Transactions on Instrumentation and Measurement*, *55*(4), 1443–1448. https://doi.org/10.1109/TIM.2006.876386.

Zhang, L., & Zhang, G. (2024). Semantic feedback for generalized zero-shot learning. In *2024 4th international conference on neural networks, information and communication engineering (NNICE)* (pp. 298–302). https://doi.org/10.1109/NNICE61279.2024.10499130.